

Book Cover



DOINGG@UNION.EDU

COURSE: CSC:483 SPECIAL TOPICS - AP-
PLIED BIOLOGICAL DATA SCIENCE
DOCUMENTATION

SELF

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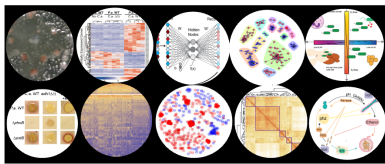
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Home



The course website for [CSC483](#), part of the Union College [CS curriculum](#). Here you can find our weekly schedule, assignments and resources.

Announcements

Upcoming due dates

Next assignment (HW 2) is due Tues Jan 20th **before** class.

Office Hours

- Student/Office Hours:
 - Steinmetz 108B
 - Wednesday 2:00 pm - 3:30 pm
 - Thursday 4:00 pm - 5:30 pm
 - subject to change, check course website for most up-to-date schedule
 - drop-in or schedule a 15 minute slot: <https://calendar.app.google/8bus6pfDvyphR9ar5>
 - by appointment for another time or over zoom

Course Description

In this course we will gain familiarity and practice in methods of analyzing biological sequence data and the large datasets that have resulted from decades of high-throughput experimentation. We will follow how data are turned into biological knowledge through the general workflow of (1) sequence alignment, (2) dimensionality reduction and (3) statistical, comparative hypothesis testing. At each stage we will compare state-of-the-art deterministic methods with emerging machine learning-based counterparts and interrogate trade-offs in performance and interpretability. We will explore the role of interdisciplinary communication in biological data science and the iterative process of program development and implementation in exploring the great biological unknowns. This course will use Python and R and introduce libraries useful in applied computer science and guidance on how different approaches are used in different contexts. No prior biological knowledge will be assumed.

Prerequisite(s): MTH 197 and a C- or higher in CSC 151. Recommended: CSC 250. MTH 199 can be substituted for MTH 197 CC: SET Note: Course can be repeated for credit under different topics. Consult with the department chair for more information.

Schedule

Weekly Schedule

Schedule

W	TOPIC	Due	Notes
1	Scope of Biological Data	HW 01	
2	Arc of Analyses	HW 02	
3	Sequence Alignment	HW 03, Project Proposal	
4	Data Wrangling	HW 04	
5	Dimensionality Reduction	Exploratory Analyses	
6	Visualization	Figure Drafts	
7	Hypothesis Testing	Results	
8	Peer Review	Peer Review	
9	Methodological Comparative Analysis	Response to Reviewers	
10	Project Presentations	Final Project & Presentation	

Resources

Rosalind Problems

- Course enrollment link: <https://rosalind.info/classes/enroll/750c73565c/>

Google Colab Pro

- Pro account sign-up: <https://colab.research.google.com/signup>
- Use your Union email

Late Token Form

- These are for project-related deadlines, not Rosalind problems
- These can be used for any reason
- These do not apply to your final deadline
- Form link: <https://forms.gle/ZcEss9oNayecL8zr7>

Python Style Guide

- PEP 8
- should be used in all code cells of Colab notebooks
- Link: <https://peps.python.org/pep-0008/>

Syllabus

Syllabus

CSC483-Special Topics - Biological Data Science Syllabus

1 CSC 483: Special Topics - Applied Biological Data Science

- Union College, Winter 2026
- Georgia Doing, PhD
- email: doingg@union.edu
- office: Steinmetz 108B

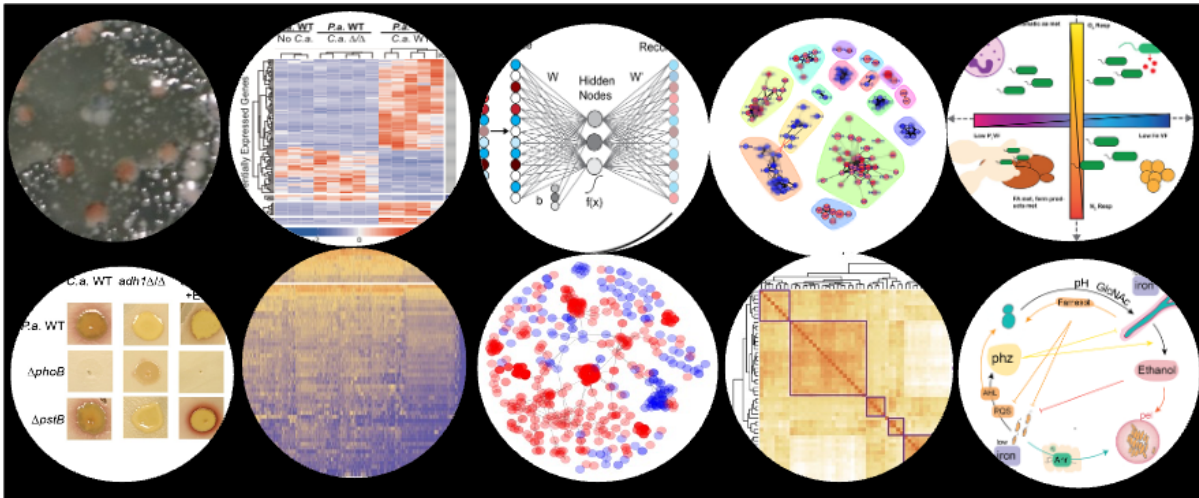


Figure 1: Views of Biological Data.

2 COURSE BASICS

- Lecture: Tuesday/Thursday 1:55 am - 3:40 pm
- Location: ISEC 070
- Student/Office Hours:
 - Wednesday 2:00 pm - 3:30 pm
 - Thursday 4:00 pm - 5:30 pm
 - subject to change, check course website for most up-to-date schedule
 - drop-in or schedule a 15 minute slot: <https://calendar.app.google/8bus6pfDvyphR9ar5>
 - by appointment for another time or over zoom

3 OVERVIEW

In this course we will gain familiarity and practice in methods of analyzing biological sequence data and the large datasets that have resulted from decades of high-throughput experimentation. We will follow how data are turned into biological knowledge through the general workflow of (1) sequence alignment, (2) dimensionality reduction and (3) statistical, comparative hypothesis testing. At each stage we will compare state-of-the-art deterministic methods with emerging machine learning-based counterparts and interrogate trade-offs in performance and interpretability. We will explore the role of interdisciplinary communication in biological data science and the iterative process of program development and implementation in exploring the great biological unknowns. This course will use Python and R and introduce libraries useful in applied computer science and guidance on how different approaches are used in different contexts. No prior biological knowledge will be assumed.

Prerequisite(s): MTH 197 and a C- or higher in CSC 151. Recommended: CSC 250. MTH 199 can be substituted for MTH 197 CC: SET Note: Course can be repeated for credit under different topics. Consult with the department chair for more information.

4 LEARNING OBJECTIVES

By the end of the course, you should be able to *answer* the following questions:}

- How is sequence data transformed into quantitative data?
- What are the fundamentals of data wrangling?
- What are best practices in data visualization?
- How can data be communicated between data scientists and biologists?

By the end of the course, you should be able to *do* the following:

- Acquire and prepare data for analysis.
- Conduct descriptive data exploration.
- Communicate findings of an independent project.

A 1-page schedule-at-a-glance is included at the end of the syllabus, but please see the course website more detailed schedule.

5 CLASS POLICIES AND GUIDELINES

The Computer Science Department as a whole welcomes all people, regardless of age, background, beliefs, ethnicity, gender identity, gender expression, national origin, religious affiliation, sexual orientation and any other differences, be they visible or non-visible. It's also important to recognize that institutional racism has prevented members of marginalized groups - especially black, indigenous and other people of color (BIPOC) - from fully participating in the field of Computer Science.

You. Yes you, the one reading this syllabus. You belong here.

As an instructor, I will do my utmost to uphold these principles and to treat everyone with respect. As students in this class, I expect you to do the same since one person is not a community unto themselves. We're all in this together, so let's treat each other well. Lastly, I welcome feedback on issues we might discuss or ways that our class can be a more just one for all people.

6 COURSE MATERIALS

Online Textbook: Runestone (free).

We are going to use an online, interactive, free and open source textbook. This textbook has a lot of embedded interactive exercises. Reading this book always also means doing the activities. To get access to the book, please register on Runestone using the following information which can also be found on Nexus:

- <https://runestone.academy/runestone/default/user/register>
- your Union College email
- unioncollege_py4e-int_fall25 as the course name

We'll use the following websites and software throughout the course:

- Rosalind problems: <https://rosalind.info/classes/enroll/750c73565c/>

- The course website: <https://georgiadoing.github.io/tufte-quarto/>
- Google Colab: the cloud-based platform to write and run Jupyter Notebooks with python code
 - Pro account sign-up: <https://colab.research.google.com/signup>
 - Use your Union email

7 COURSE RESOURCES

The course website has a list and links to the readings and software resources that we are going to use. In class, we are going to use the Linux machines in ISEC 070. Outside of class you have a choice of hardware. You can access Google Colab via a browser on your own machine, install the software on your own computer (Python is freely available for Windows, Mac, and Linux) or you can work in one of the Computer Science labs. We have three spaces that you can use 24/7 using your ID card, except when classes are being held in them:

- Olin 107
- PASTA Lab (ISEC 051+)
- Computer Science Resource room (Steinmetz Hall, 209A)

8 LIBRARY RESOURCES

The library is available to help you with your research needs! Librarians can help you develop research questions, search for and select the best sources for your projects, identify research strategies, evaluate sources, and assist you with creating citations. There are multiple ways for you to contact a librarian. For more information, please see our Ask A Librarian page: <https://www.union.edu/schaffer-library/ask-a-librarian>.

9 ACCOMODATIONS

It is the policy of Union College to make reasonable accommodations for qualified individuals with disabilities. If you are a person with a disability and wish to request accommodations to complete your course requirements, please make an appointment with me or stop by during my office hours as soon as possible, all discussions will remain confidential. You must provide reasonable notice and be in touch with Accomodative Services on the 2nd floor of Schaffer Library, if you have not already, if you wish to take advantage of extra time on exams.

10 GRADING

The following components will contribute to your final grade for this class with roughly the indicated weight.

- Final project: 25%
- Final oral exam: 25%
- Problem sets: 25%
- Peer review: 25%

Letter Grade Scale:

- A: 93-100; A-: 90-92
- B+: 87-89; B: 83-86; B-: 80-82
- C+: 77-79; C: 73-76; C-: 70-72
- D: 60-69; F: 0-59

11 LATE POLICIES

Homework assignments will usually be discussed in class on the day that they are due and thus, without prior agreement, there is NO late submission of assignments allowed. You have three “extension tokens” that can be used for due dates related to your final project for this class. Each token will extend a single project deadline by 24 hours. You can use each token on a different deadline, or use two or even all three on a single deadline. Once all three tokens have been used, no late submissions will be accepted (barring exceptional circumstances).

<https://forms.gle/ZcEss9oNayecL8zr7>

12 ATTENDANCE

Class participation is a critical component of the course and attendance is mandatory. I realize that sometimes other things come up (interviews, athletic events, illness, etc.). In those cases, just let me know (in advance, if at all possible) that you are going to be absent. You may not receive credit for make-up material if you did not discuss your absence with me prior to class. No matter how much class you have missed, please do not come to class if you have an illness that is likely contagious, please email me to work out a reasonable solution.

If you do miss class, it is your responsibility to make up any material that you missed. Get notes from a classmate, make sure to complete all homework assignments due or assigned

during the class that you missed, and come see me if you have any questions on the material. Unless you have made an arrangement with me ahead of time or you had an emergency, I will expect you to hand in all assignments by the due date, regardless of whether you were in class. You will not be able to make up missed exams or in-class questions and assignments.

Attendance in class constitutes being present, respectful and engaged. Accessing off-topic websites or software, checking email or phones, *utilizing large language models* or otherwise attending to things not pertinent to the course is distracting to yourself and others and will result in a reduction of your overall grade, up to a 0.5% reduction per class.

13 PARTICIPATION AND ENGAGEMENT

Following these guidelines will constitute positive engagement:

- Respect. This course is a space for rigorous and respectful debate. When confronting ideas and people different from you, lead with curiosity rather than judgement. This commitment extends to all our face to face and digital interactions. We will critique ideas, not people. To protect our shared learning environment, you may not record, photograph, or share any part of our class sessions outside of our class.
- Show up to class on time and prepared. Show up ready to learn by completing the readings and exercises. When you are late or unprepared, you are disrespecting the learning experience for the group.
- Embrace intellectual humility. Recognize that there are limitations to your knowledge and that some of your beliefs could actually be wrong. Be curious about your thinking and open to learning from others. This is hard, but so vital to your learning in this course—we'll work on developing this throughout the term!
- Get help when you need it. If you are stuck or confused or lost, be proactive and get help. The best learners and thinkers can figure out when they are stuck and make a plan to get unstuck. Come to Office Hours (Wed/Thurs 2:00-3:30pm Steinmetz 108B), the CS Helpdesk (Sun-Thurs 7:00-9:00pm Olin 107) or ask another student. Often the best person to explain something is the person who just figured it out. Try reaching out to another student in the course who seems like they get it—they will probably be flattered you asked!

14 ACADEMIC INTEGRITY AND "ARTIFICIAL INTELLIGENCE" USE

Union College recognizes the need to create an environment of mutual trust as part of its educational mission. Responsible participation in an academic community requires respect for and acknowledgement of the thoughts and work of others, whether expressed in the

present or in some distant time and place. Matriculation at the College is taken to signify implicit agreement with the Academic Honor Code, available at: <http://muse.union.edu/honorcode>

It is each student's responsibility to ensure that submitted work is their own and does not involve any form of academic misconduct. Students are expected to ask their course instructors for clarification regarding, but not limited to, collaboration, citations, and plagiarism. Ignorance is not an excuse for breaching academic integrity. Students are also required to affix the full Honor Code Affirmation, or the following shortened version, on each item of coursework submitted for grading:

Union College has an honor code as follows. That material will appear in every notebook you submit to me for grading.

As a student at Union College, I am part of a community that values intellectual effort, curiosity and discovery. I understand that in order to truly claim my educational and academic achievements, I am obligated to act with academic integrity. Therefore, I affirm that I will carry out my academic endeavors with full academic honesty, and I rely on my fellow students to do the same.

Scholastic dishonesty is misrepresenting someone else's work as your own and will not be tolerated.

For this class in particular, we encourage working together during class time. If you missed something, talk to me, talk to your classmates or reach out via piazza or helpdesk. However ALL HOMEWORK must be completed individually. You are encouraged to:

- Talk about concepts in solutions
- Discuss ideas
- Look up online documentation, or examples

You MAY NOT:

- Share code
- Look at another student's code
- Look up solutions to specific problems on the internet
- Copy or paste any text or code to or from a large language model

Ultimately, I may choose to call you into office hours to explain the choices you made in solving a problem. You should be comfortable with explaining the choices you made, how that choice was implemented and why.

Your goal for discussions about any assignment should be that you come away with a better understanding of the problem and of possible ways to approach it so that you can then try out these approaches on your own. You should never leave a discussion with just an answer, without an understanding of how to arrive at that answer. A good general guideline for any

discussion, or interaction with an AI model, is that you should not leave the discussion with anything written or typed.

15 CONTACTING ME OUTSIDE OF CLASS

The best method for contacting me outside of class is to stop by my office during Office Hours or whenever my office door is open. You can also schedule a time with me if my regular office hours don't work. Please contact me via email (doinggg@union.edu) and we can arrange a phone or zoom call. I respond to emails as soon as possible, but it may take up to one day to get a response during the term, and longer between terms.

16 ADDITIONAL RESOURCES

Mental Health and Campus Resources

Union College is committed to supporting and advancing the mental health and well-being of our students. During the course of their academic careers, students often experience personal challenges that contribute to barriers in learning, such as drug/alcohol problems, strained relationships, chronic worrying, persistent sadness or loss of interest in enjoyable activities, family conflict, grief and loss, domestic violence, difficulty concentrating, problems with organization, procrastination and/or lack of motivation. Students also sometimes come to college with a history of learning difficulties (e.g., any form of special education), experience difficulties succeeding in a particular subject (e.g., math, reading), or have experienced some form of trauma be it emotional or physical (e.g., head injury). These mental health concerns can lead to diminished academic performance and can interfere with daily life activities. If you or someone you know has a history of mental health concerns or if you are unsure and would like a consultation, a variety of confidential services are available. The Eppler-Wolff Counseling Center provides free counseling and therapy services (including psychiatry) to all enrolled students. Please call (518) 388-6161 or visit the Wicker Wellness Center in person any weekday between 8:30 and 5:00 to schedule an initial contact appointment. Visit the Counseling Center website for more information.

In a crisis situation, or after hours, contact Campus Safety at (518) 388-6911. The National Suicide Prevention hotline also offers a 24-hour hotline at (800) 273-8255.

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students (dos_office@union.edu) for support or drop into office hours Monday - Friday 8:30 am - 5:00 pm in the Reamer Campus Center room 306. Furthermore, please notify me if you are comfortable doing so and I will provide any resources I can. The Union College Persistence Fund can provide financial support in times of unexpected hardship to cover

needs such as emergency medical expenses not covered by insurance and basic living expenses. Additional sources of support are outlined on the Dean of Students webpage: <https://www.union.edu/dean-students> .

16.1 ADDENDA

Any community agreed upon additions:

17 TENTATIVE SCHEDULE

W	TOPIC	Due	Notes
1	Scope of Biological Data	HW 01	
2	Arc of Analyses	HW 02	
3	Sequence Alignment	HW 03, Project Proposal	
4	Data Wrangling	HW 04	
5	Dimensionality Reduction	Exploratory Analyses	
6	Visualization	Figure Drafts	
7	Hypothesis Testing	Results	
8	Peer Review	Peer Review	
9	Methodological Comparative Analysis	Response to Reviewers	
10	Project Presentations	Final Project & Presentation	

18 PROGRESS TRACKER

	Week	Week	Week	Week	Week	Week	Week	Week	Week	Week
Learning Goals	1	2	3	4	5	6	7	8	9	10
How is sequence data transformed into quantitative data										
Data wrangling										
Data visualization										
Interdisciplinary communication										
Aquire and prepare data										
Descriptive analysis										
Independent project										

Grading

Table 2: Grading Scheme

Course Element	Grade Point Contribution	Notes
Final Project	25%	written
Final Presentation	25%	
Problem Sets	25%	written
Peer Review	25%	written
total	100%	

Table 3: Letter Grade Scale

Letter Grade	Percent range
A	93 - 100
A-	90 - 92
B+	87 - 89
B	83 - 86
B-	80 - 82
C+	77 - 79
C	73 - 76
C-	70 - 72
D	60 - 69
F	0 - 59

Assignments

CSC483 HW01

CSC 483: Homework 1, Due Jan 13, 2026

Rosalind problems 1 - 6: A review of Python.

1. Setup Colab Notebook (1 pts)

1. Make a copy of the Colab Notebook template:

https://colab.research.google.com/drive/1PdhQTAnEY_IPhAQkBxZcACU0gJoOp6oP?usp=sharing

2. Rename it with your own name: [Name]_CSC483_template.ipynb
3. Fill your name into the header. For now, leave the date and assignment name generic.
4. Move this notebook to your class folder on Google Drive, in a well-organized directory.
5. Share it with me, doingg@union.edu
6. Now make a copy of your edited template, save it to your drive, and rename it: [Name]_CSC483_HW01.ipynb
7. You will also share your HW01 notebook with me as your submission.

2. Rosalind Problems (1 pts)

Enroll in our class on Rosalind: <https://rosalind.info/classes/enroll/750c73565c/>

Be sure to use your Union email for your account.

You should see the first assignment (problems 1-6).

Format

For each Rosalind problem, use chunks of Markdown text and python code to work out your solution.

Each Rosalind problem will give you a sample data and output to work on. Then, when you are ready, you will download a new dataset and past the corresponding result into the prompt box on Rosalind. I am not asking you to submit code to Rosalind as you will save it in your Colab notebook.

Note on reading/writing files

For some problems you may need to read or write files. In these cases, mount your google drive and upload/save files there.

```
from google.colab import drive
drive.mount('/content/drive/')
```

To check which directory your notebook is mounted to, you can use bash within python chunks

```
!ls
```

You can then use the appropriate relative paths to your files of interest. For example, I used:

```
f = open("drive/MyDrive/Teaching/CSC483-W26/test_text.txt", "r")
```

3. Setup Check-in Notebook (1 pts)

1. Make a copy of the Check-in Notebook template:

<https://colab.research.google.com/drive/1e4a0xjIn9oUVCwZNcZrkv3lM4iyDfSxK?usp=sharing>

2. Rename it with your own name: [Name]_CSC483_check-in_template.ipynb

3. Follow the instructions in the Colab Notebook.
4. Share your notebook with me (doingg@union.edu) as your submission.

4. *Office Hours (1 pts)*

1. After completing your check-in notebook, come to office hours some-time in the next 2 weeks.
 - Location: ISEC 070
 - Student/Office Hours:
 - Wednesday 2:00 pm - 3:30 pm
 - Thursday 4:00 pm - 5:30 pm
 - subject to change, check course website for most up-to-date schedule
 - drop-in or schedule a 15 minute slot: <https://calendar.app.google/8bus6pfDvyphR9ar5>
 - by appointment for another time or over zoom
 - Be careful not to procrastinate, 2 classes worth of students share the office hours.

CSC483 HW02

CSC 483: Homework 2, Due Jan 20, 2026, at 1 pm (with time for me to make copies before class)

1. Rosalind problems 7 - 13: working with sequence data. (2 pts)

1. Make a copy of your homework colab template, save it to your drive, and rename it: [Name]_CSC483_HW02.ipynb
2. You will also share your HW02 notebook with me as your submission.
3. Complete Rosalind problems 7 through 13 in the same format as you used for HW 1.

For each Rosalind problem, use chunks of Markdown text and python code to work out your solution.

Each Rosalind problem will give you a sample data and output to work on. Then, when you are ready, you will download a new dataset and past the corresponding result into the prompt box on Rosalind. I am not asking you to submit code to Rosalind as you will save it in your Colab notebook.

Work on activity: plotting metadata (1 pts)

We'll have 30 mins in class on Tuesday before looking at each others plots, so spend some time working on the metadata notebook if you will need more time than that.

Your grade point will come from being ready to present/share after the 30 mins of class time.

1. Also in Colab, keep working with the metadata introduced in class.
2. Make at least 3 plots analyzing or summarizing columns.
3. Write down (in Markdown) a comparison you think would be interesting to make, why you think it would be interesting and what you hypothesize the result would look like.
 - you don't have to succeed in your comparison yet, if you'd need more tools/libraries/functions to do so
 - it is more important that you have a reason for making the comparison and an estimate to which you could compare your results

3. Update Check-in Notebook (1 pts)

1. Review your check-in notebook and scores for last week's homework assignment
2. Include a Markdown chunk that attests to my notes or requests a correction/update

CSC483 HW03

CSC 483: Homework 3, Due Jan 27, 2026, at 1 pm (with time for me to make copies before class)

1. Rosalind problems 14 - 20: sequence & alignment. (3 pts + 4 pts of PR)

1. Make a copy of your homework colab template, save it to your drive, and rename it: [Name]_CSC483_HW03.ipynb
2. You will also share your HW03 notebook with me as your submission.
3. Complete Rosalind problems 14 through 20 in the same format as you used for HW 1.

Note that for this set, some problems direct you to use web-based programs rather than writing your own python code.

For each Rosalind problem, use chunks of Markdown text and python code to work out your solution.

Each Rosalind problem will give you a sample data and output to work on. Then, when you are ready, you will download a new dataset and past the corresponding result into the prompt box on Rosalind. I am not asking you to submit code to Rosalind as you will save it in your Colab notebook.

4. In-class peer review holds an additional 4 points in the grading scheme, so be sure to finesse your notebook for easy reading.

2. Update Check-in Notebook (1 pts)

1. Move your reflections from HW1, HW2 and HW3 into your check-in notebook.
2. Review your check-in notebook and scores for last week's homework assignment
3. Include a Markdown chunk that attests to my notes or requests a correction/update

Notes

Week 1

Week 1 slides

**CSC 483 - Applied Biological Data Science -
W1D1**

Today (105 min)

- Introductions? (15')
- Fermi questions (30')
- What is biological data science? (30')
- Course goals, arc, project (15')
- Colab rules & grading scheme (15')

Free write: scale of “biology” to “data science”

- what classes or experiences have you had? where do they fall on the spectrum?
- where would you place yourself on the spectrum?
- in which direction do you hope to move by taking this class?

Me:

- Georgia Doing, PhD
- she/her/hers
- computational microbiologist studying human skin microbiome and uncharacterized genes
- interests: computational microbiology, small cells & big data

My dog Ginny



* Please email me ahead of office hours if you would like me to make sure Ginny won't be there, not a problem at all!

You:

- Name and pronouns
- Help me find you on my roster
- What is something you bring to this class? What classes or experiences have you had? Where would you place yourself on the spectrum from biology to data science?
- What is something you hope to learn in this class? In which direction (toward biology or data science) do you hope to move by taking this class?

Fermi questions

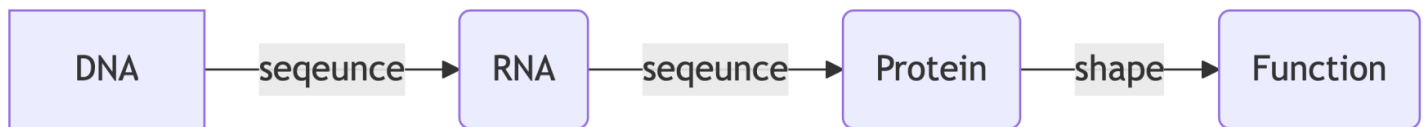
<https://fermi-questions.andrecheek.com/>

- in pairs
- if you want a different question, refresh page
- 10 mins
- answers are in orders of magnitude

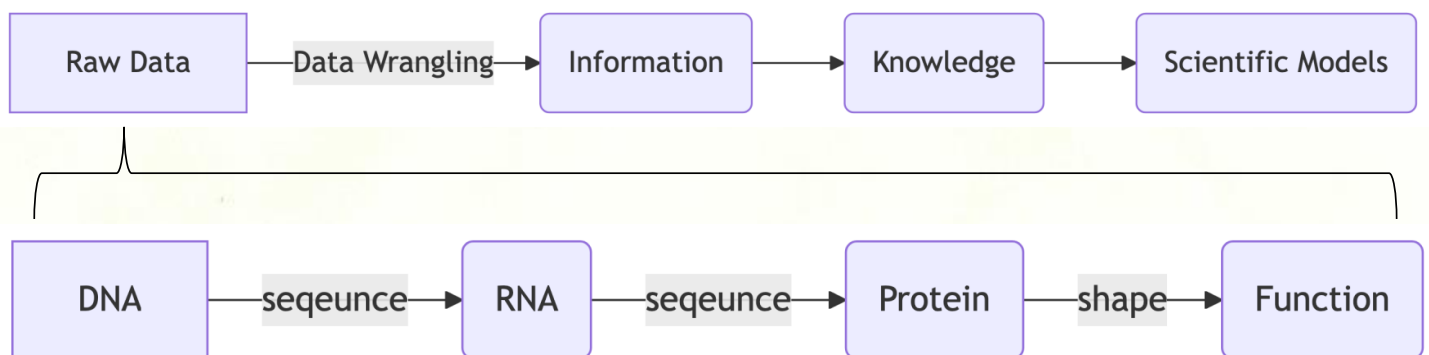
What is Data Science?



What is Biological Data?



What is Biological Data Science?

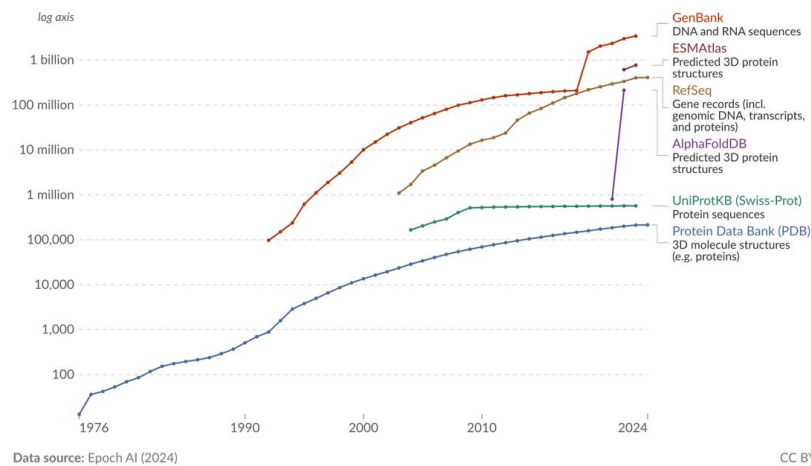


Biological Data Accumulation

Number of entries in biological sequence databases

Biological sequence databases store data such as DNA, RNA, and amino acid sequences and 3D protein structures. This data includes entries from GenBank, RefSeq, PDB, UniProtKB/SwissProt, as well as predicted protein structures in AlphaFoldDB and ESMAtlas.

Our World
in Data



<https://ourworldindata.org/grapher/number-of-entries-in-biological-sequence-databases>

What is Biological Data Science?

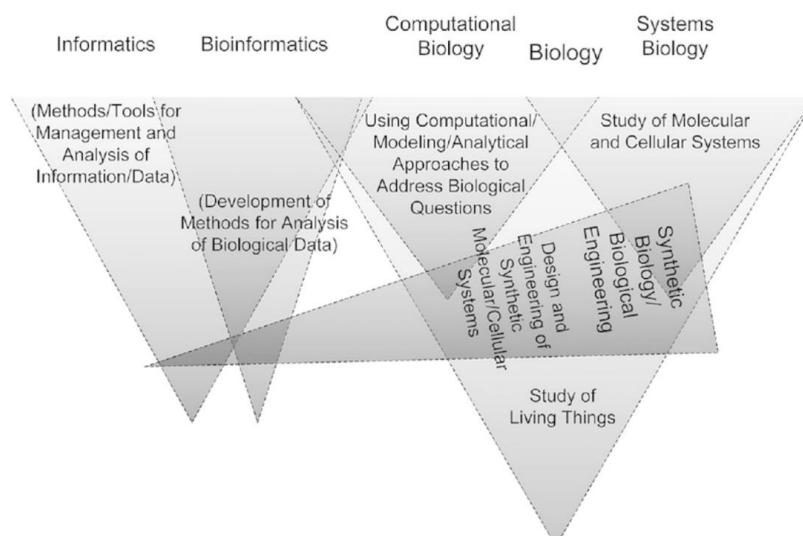


FIGURE 2.1 A disciplinary map of “informatics,” “bioinformatics,” “computational biology,” “biology,” and “systems biology” according to one practitioner. (Christopher Burge, “Introduction to Computational and Systems Biology” (7.91), lecture, Massachusetts Institute of Technology, February 8, 2007. Reproduced with permission.)

What is Biological Data Science?

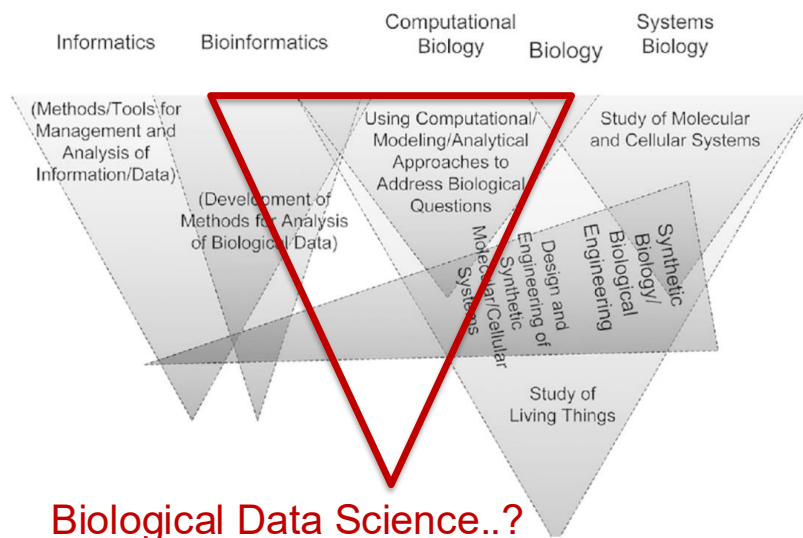
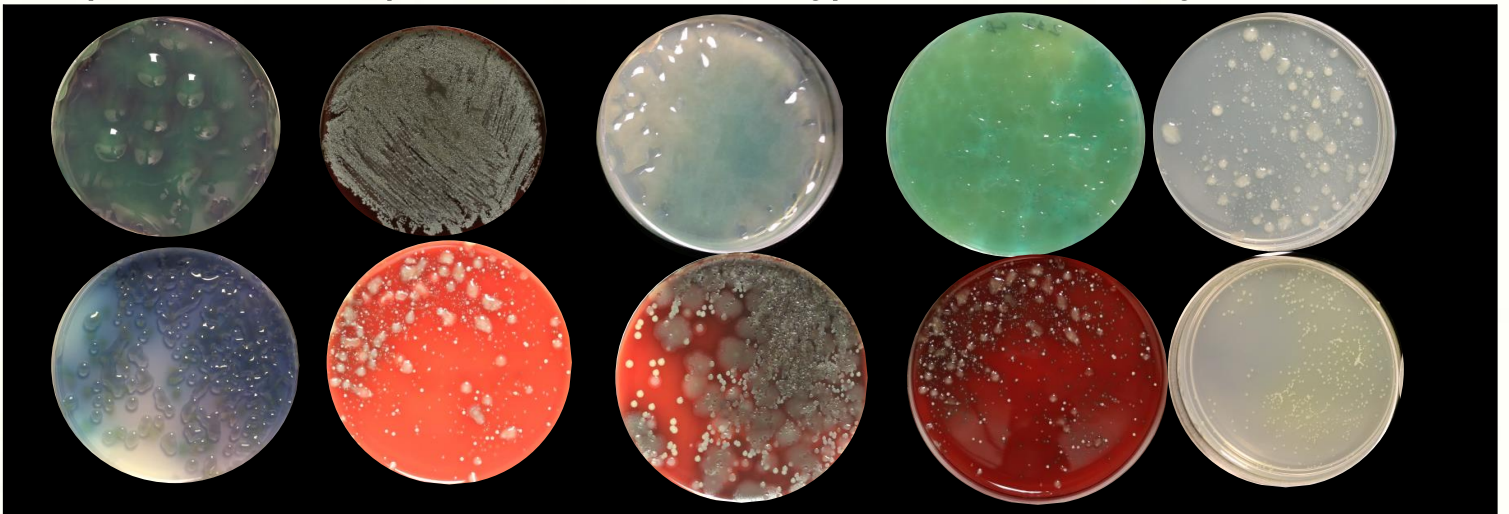


FIGURE 2.1 A disciplinary map of “informatics,” “bioinformatics,” “computational biology,” “biology,” and “systems biology” according to one practitioner. (Christopher Burge, “Introduction to Computational and Systems Biology” (7.91), lecture, Massachusetts Institute of Technology, February 8, 2007. Reproduced with permission.)

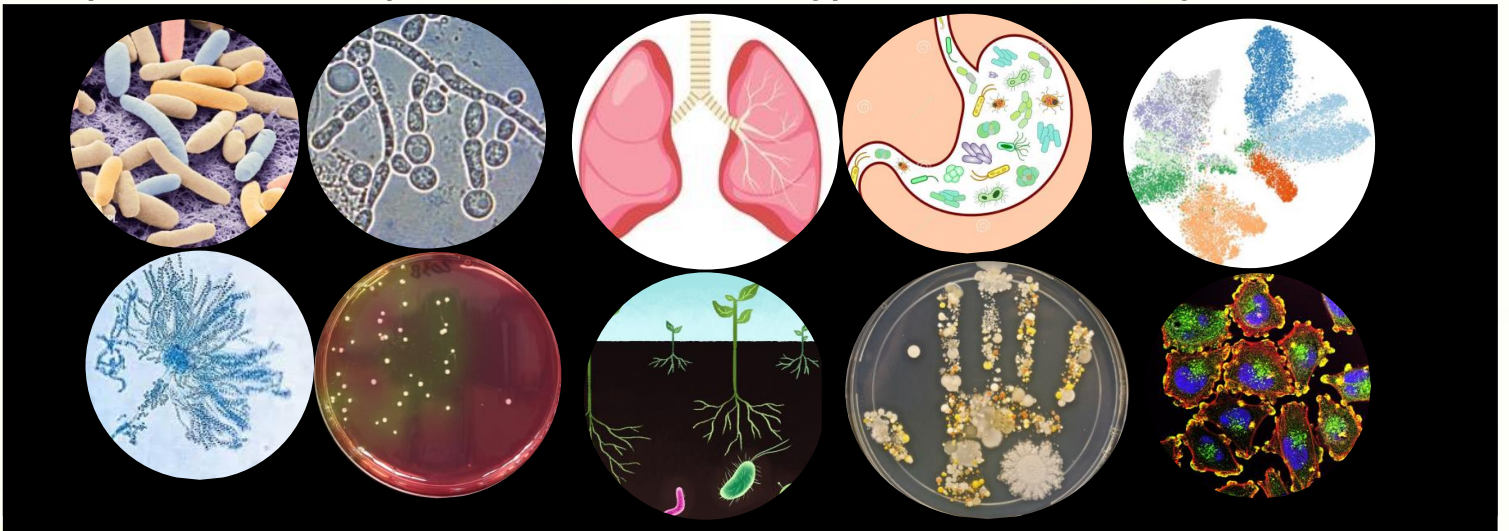
Biological Data Science to Me

My interests: computational microbiology, small cells & big data



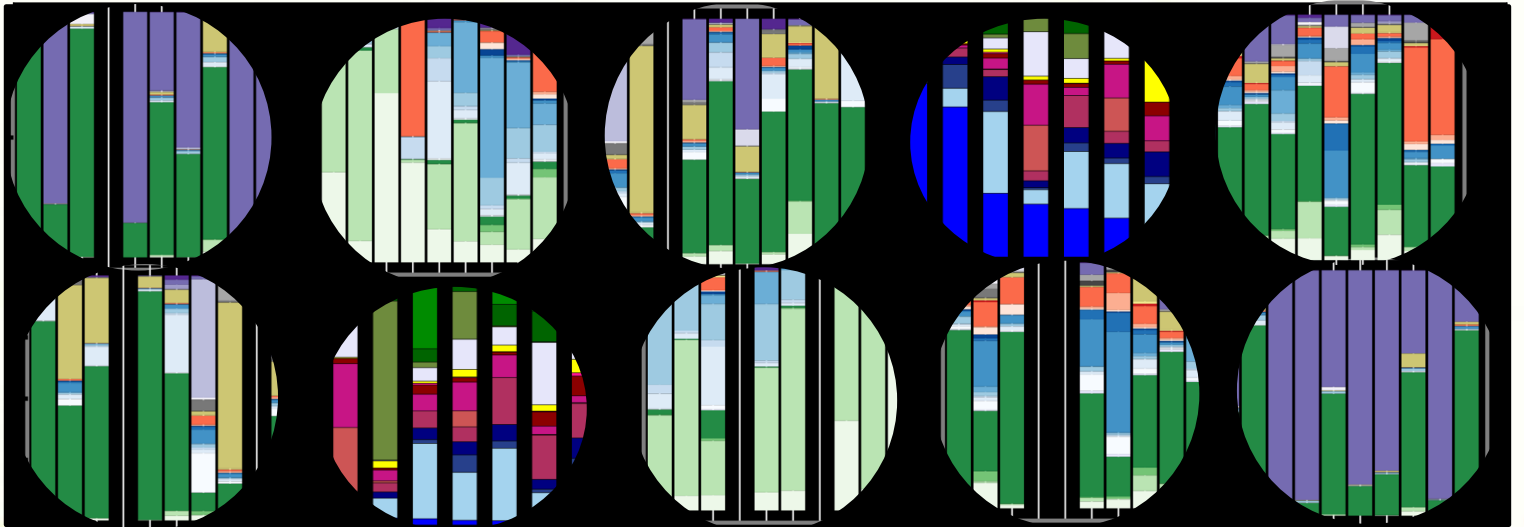
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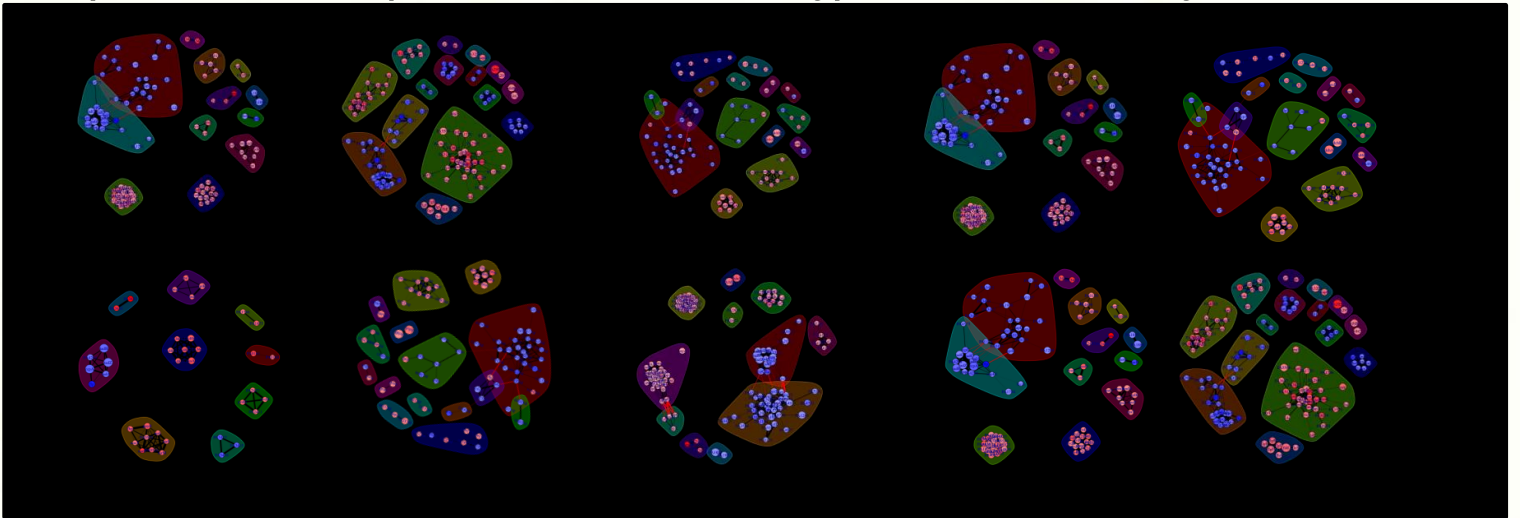
Biological Data Science to Me

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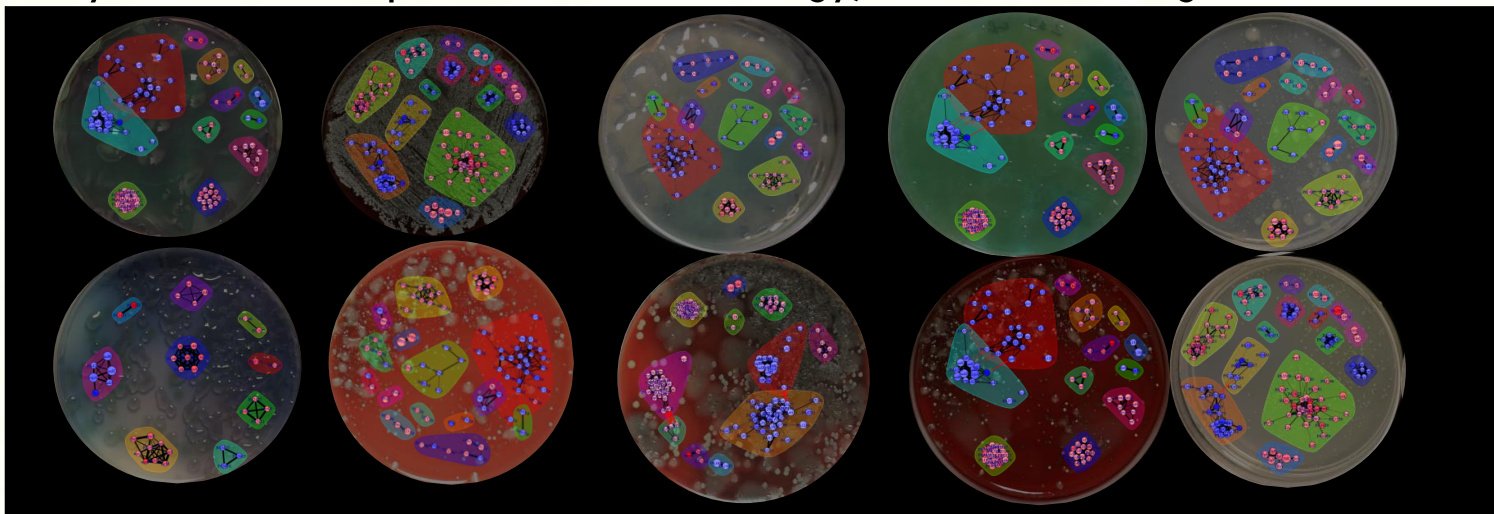
Biological Data Science to Me

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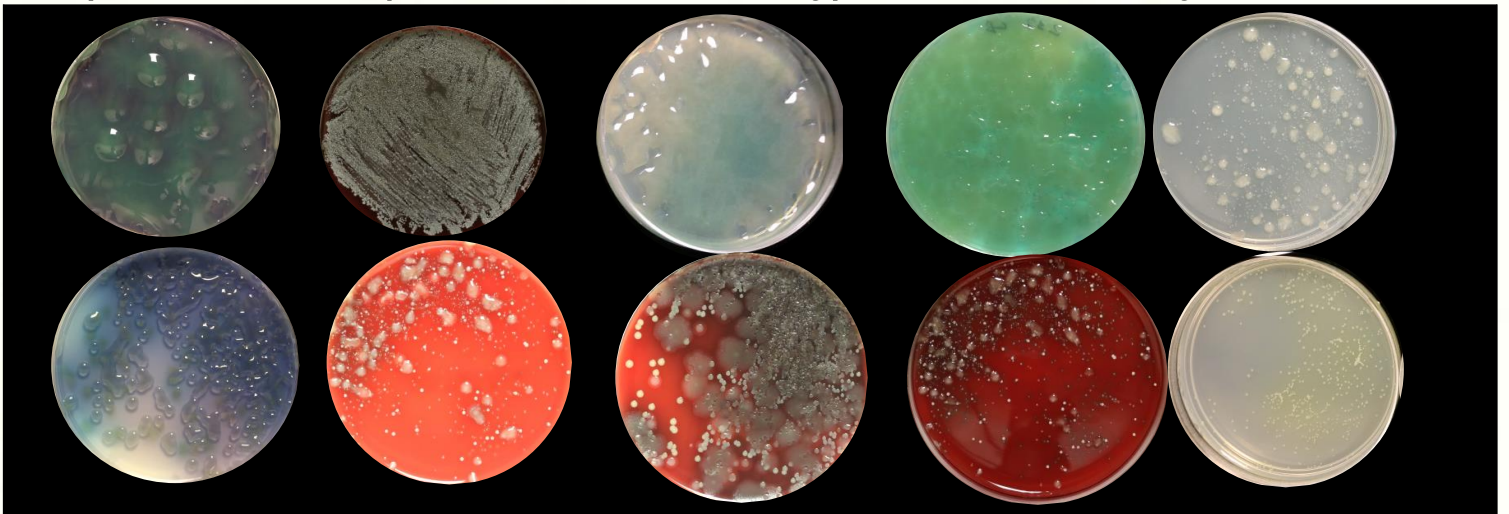
Biological Data Science to Me

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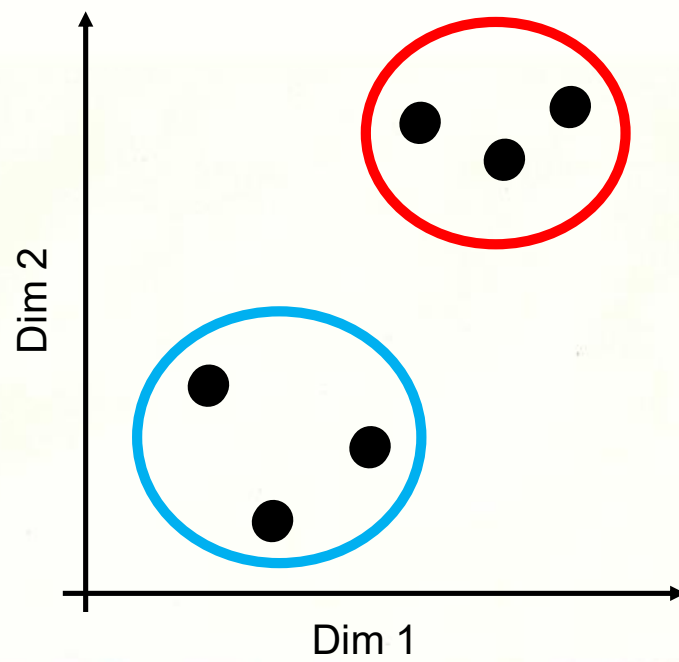
Biological Data Science to Me

5,549 Genes

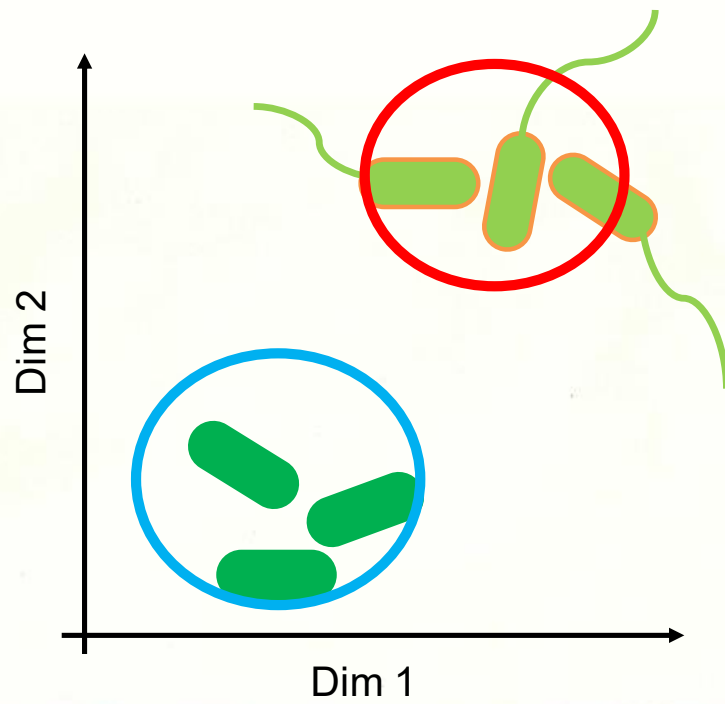
3,011 Samples



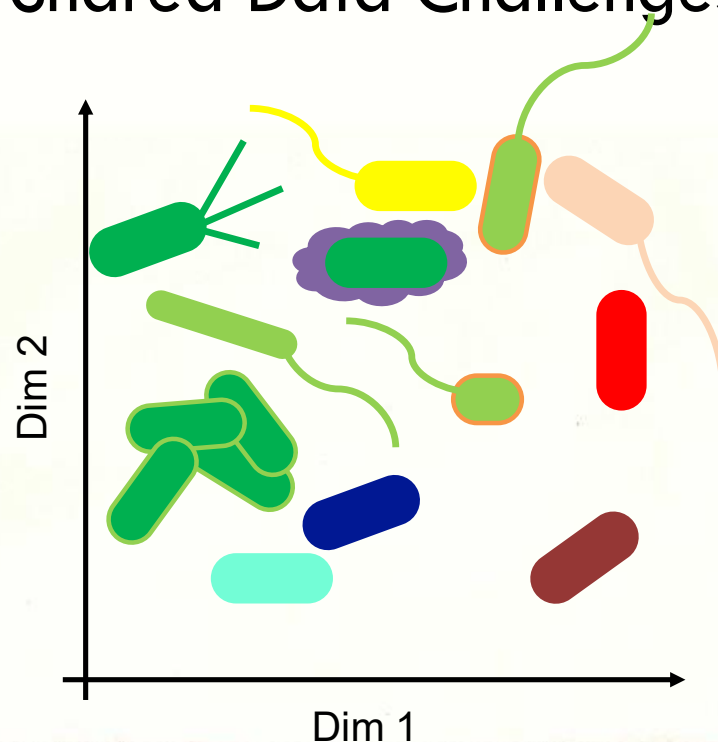
Generic Data Problems



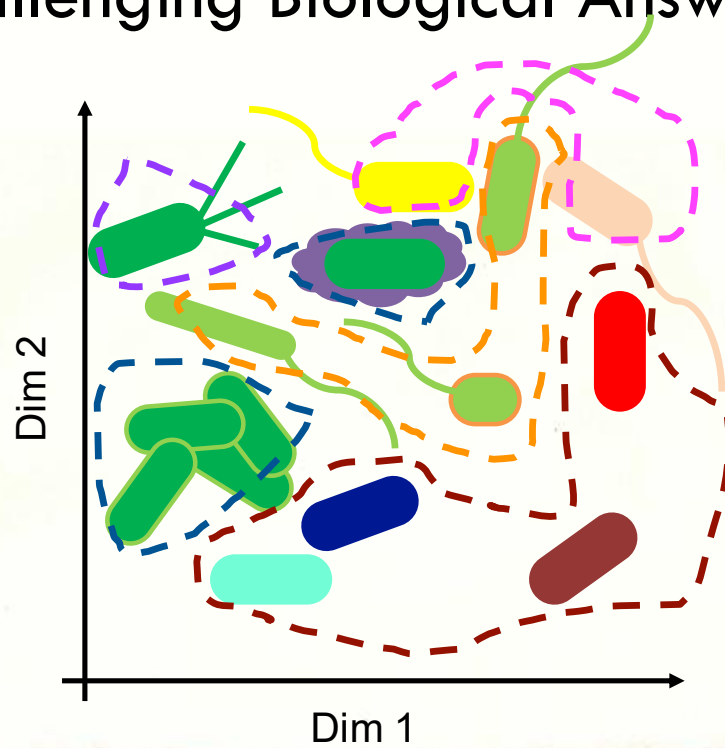
Specific Biological Questions



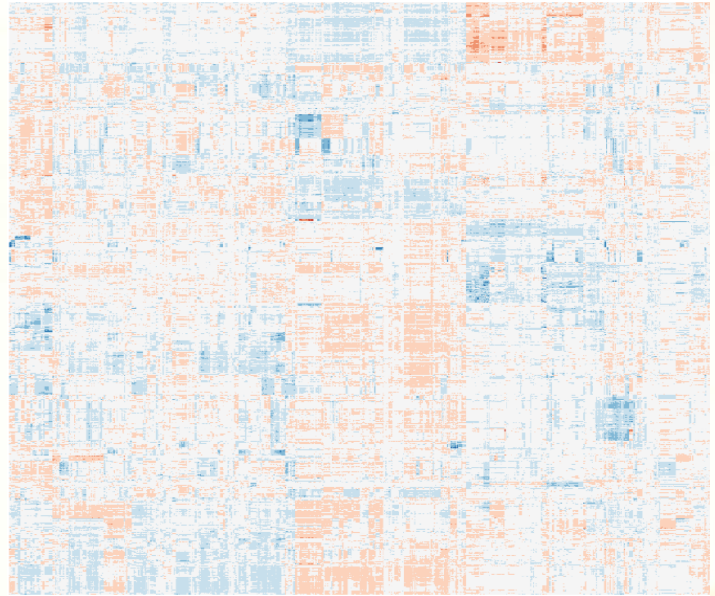
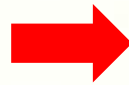
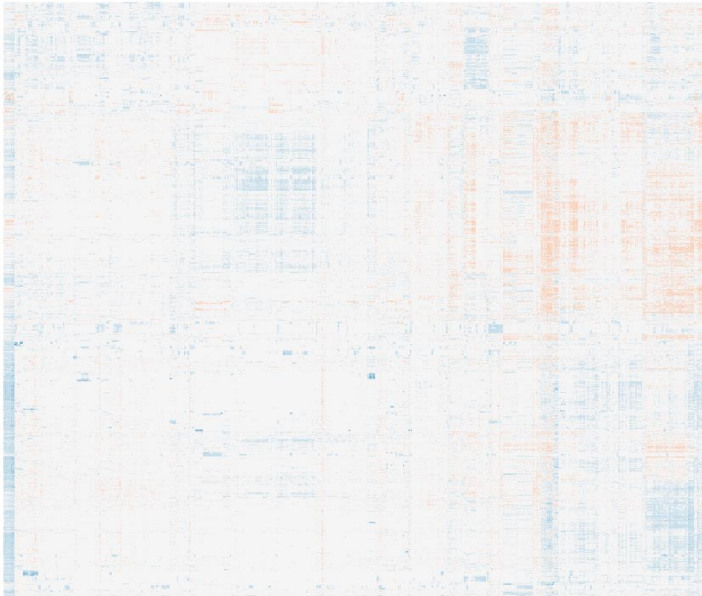
Shared Data Challenges



Challenging Biological Answers



State-of-the-Art ... Open Questions



Lab machines: getting started

- Log on:
 - Your Union usernames
 - Default password: the word union followed by your ID number; e.g., union12345

Always log off before leaving the lab

Week 2

E coli metadata activity

- colab notebook: https://colab.research.google.com/drive/1X18-sF7Spht9ZXZlDkY2S_zvgEEf4xJG?usp=sharing
- metadata 1: https://drive.google.com/file/d/1BB7M3eWo8zPuxmlwSKMzE_RBeYrjv1n0/view?usp=sharing
- metadata 2: https://drive.google.com/file/d/18_PkR_sOLsQshSM4O1n2hVOcZoxgn8E0/view?usp=sharing
- metadata 3: https://drive.google.com/file/d/1_cJGV1gwHXzgUe7DMwHVI28nhJcma2tt/view?usp=sharing

Week 2 slides

CSC 483 - Applied Biological Data Science
W2D2

CSC 483: Applied Biological Data Science

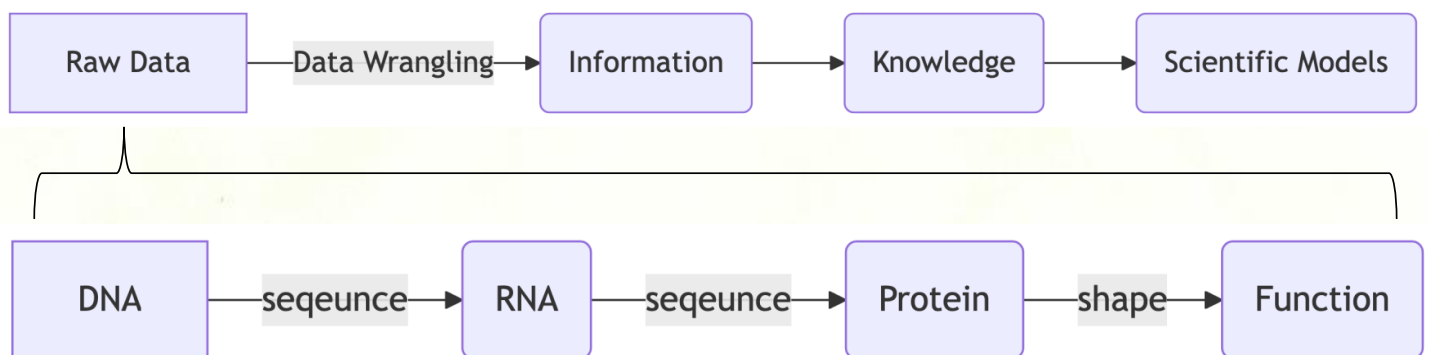
- Georgia Doing, PhD (she/her/hers)
- email: doingg@union.edu
- office: Steinmetz 108B
 - office hours:
 - Wednesday 2:00-3:30 pm
 - Thursday 4:00-5:30 pm

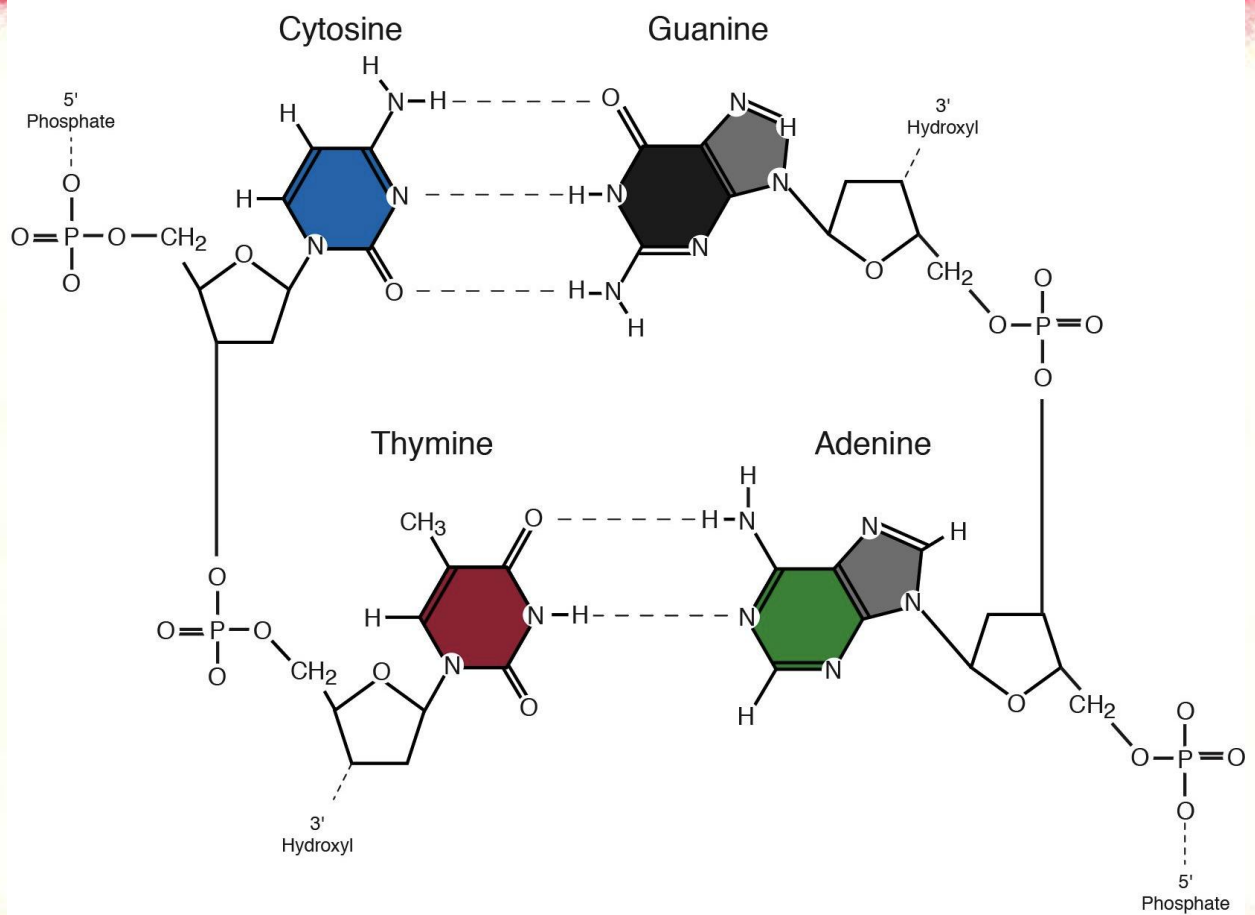
Feel free to drop in whenever my office door is open or schedule an appointment.

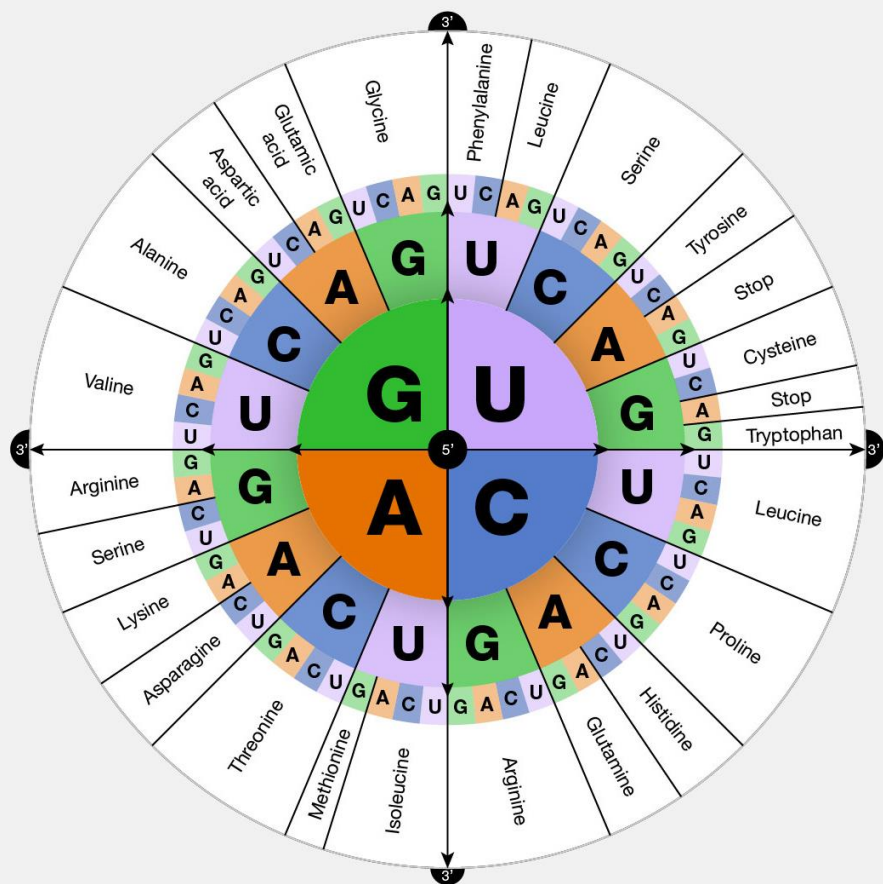
Today (145 min)

- Paper on methods/background
- Metadata summaries
- Explore data ,brainstorm ideas
- HW:
 - rosalind problems on sequence analysis (w/ colab notebooks)
 - due: tues

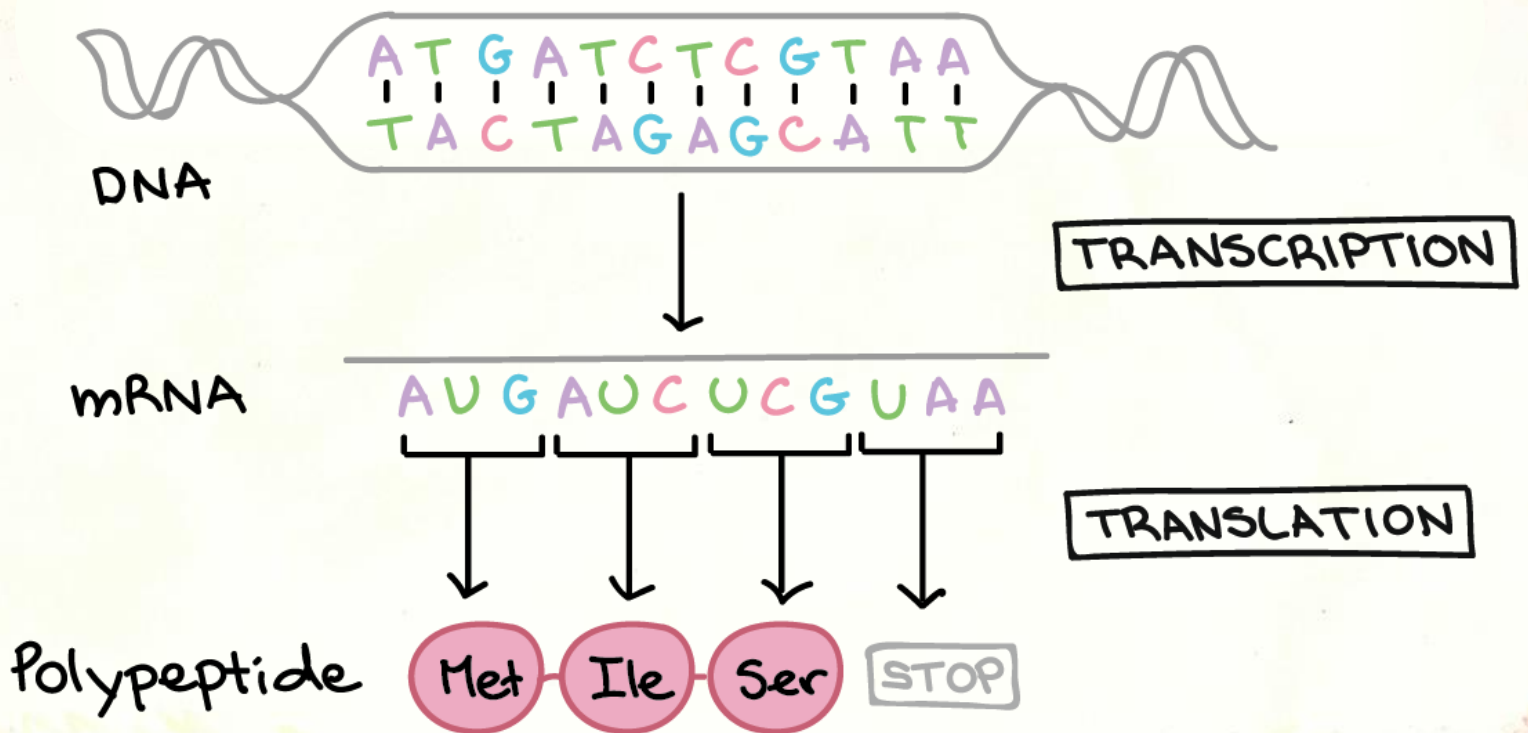
What is Biological Data Science?



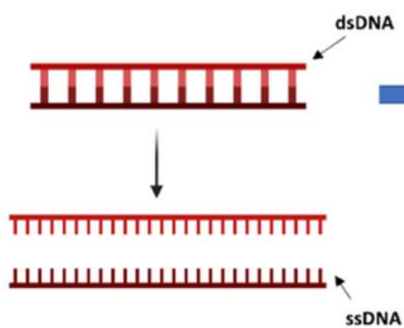




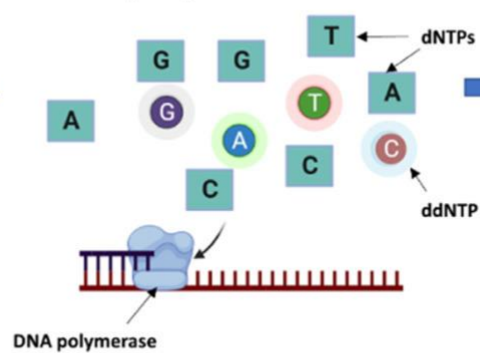
THE CENTRAL DOGMA



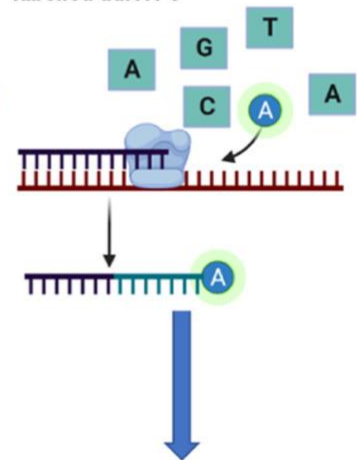
- ① Denaturation of dsDNA into ssDNA template through heat treatment



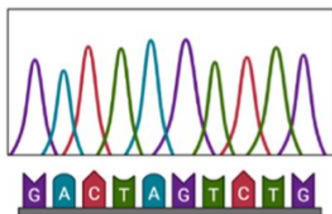
- ② Primer annealing and extension by the DNA polymerase by addition of contemporary dNTPs



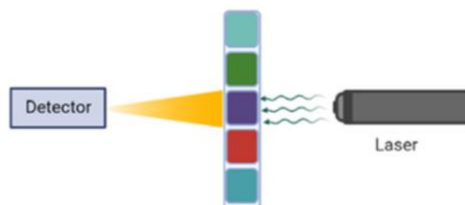
- ③ Termination of each round of primer extension by the fluorescently labelled ddNTPs



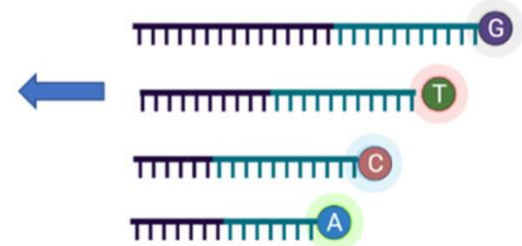
- ⑥ Sequence analysis and reconstruction



- ⑤ Separation of chain-terminated oligonucleotides using gel electrophoresis, preferably single capillary gel electrophoresis



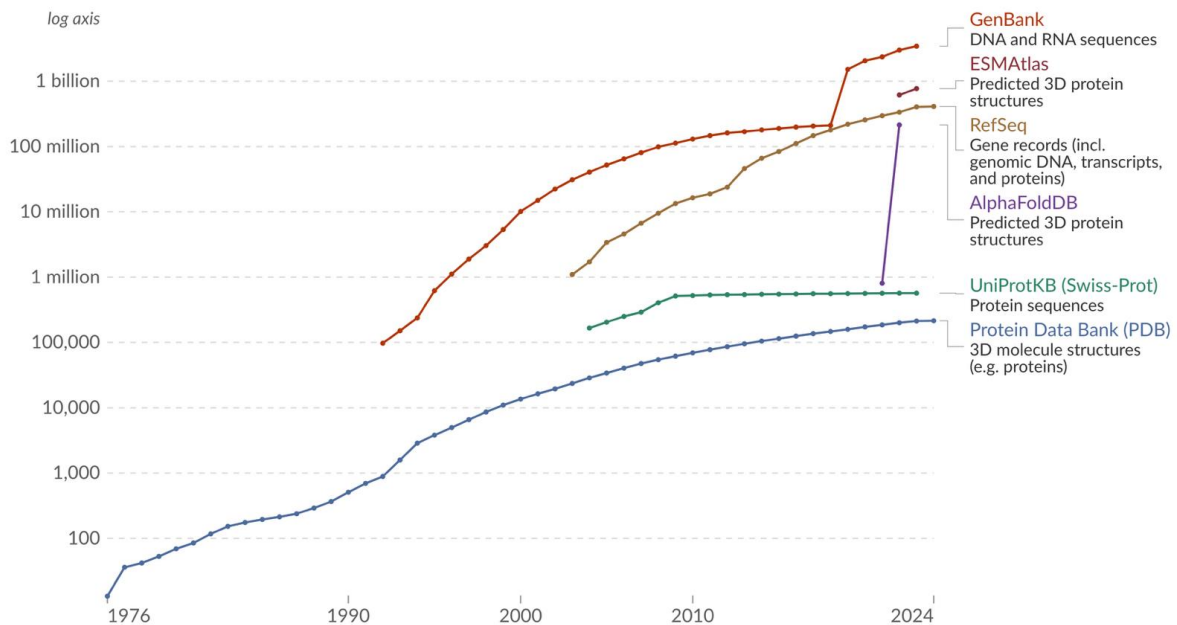
- ④ Fluorescently labelled DNA sample



Number of entries in biological sequence databases

Our World
in Data

Biological sequence databases store data such as DNA, RNA, and amino acid sequences and 3D protein structures. This data includes entries from GenBank, RefSeq, PDB, UniProtKB/SwissProt, as well as predicted protein structures in AlphaFoldDB and ESMAtlas.



Data source: Epoch AI (2024)

CC BY

<https://ourworldindata.org/grapher/number-of-entries-in-biological-sequence-databases>

Supporting open human variation data

Search IGSR

The 1000 Genomes Project created a catalogue of common human genetic variation, using openly consented samples from people who declared themselves to be healthy. The reference data resources generated by the project remain heavily used by the biomedical science community.

A world map with various regions highlighted in green and labeled with numbers. The regions include North America, South America, Europe, Asia, and Australia. The map is used to illustrate the global distribution of data sets available in the IGSR data portal.

Explore the data sets in IGSR through our data portal

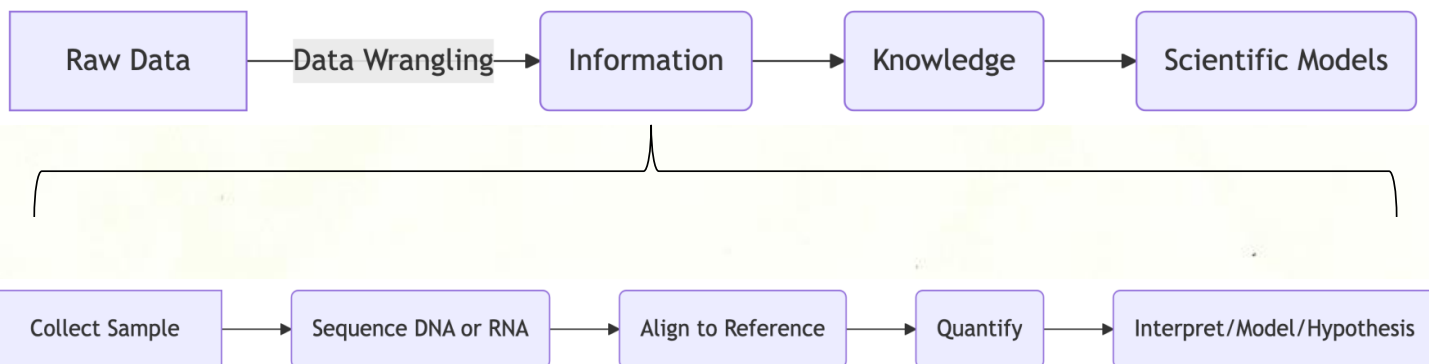
¹ Human Genome Structural Variation Consortium (HGSVC), funded by NHGRI, have built on their earlier work published in 2019 and 2021 exploring multiple technologies for structural variant discovery and short-read data generated by HGSVC.

Access HGSC data

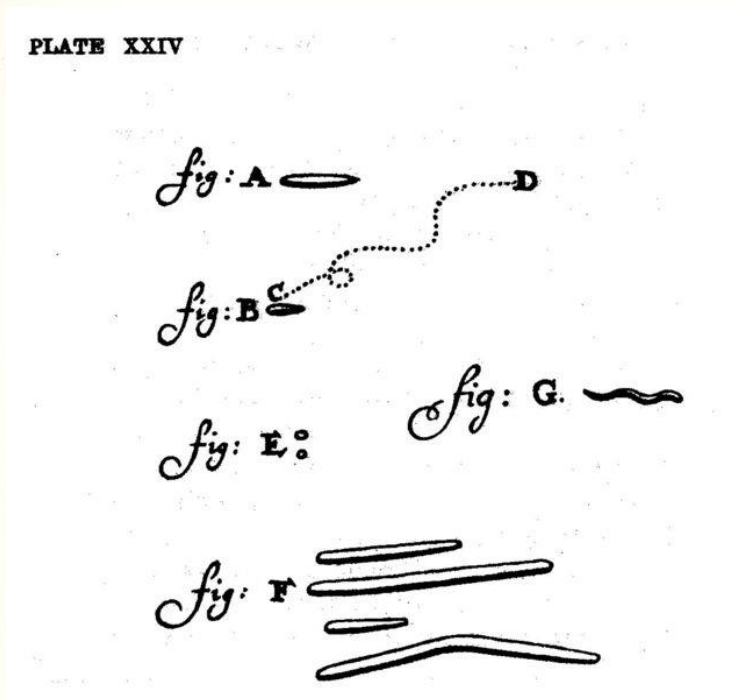
View variants in genomic context
in Ensembl

1000s Genomes Project: <https://www.internationalgenome.org/>

Experimental Workflow

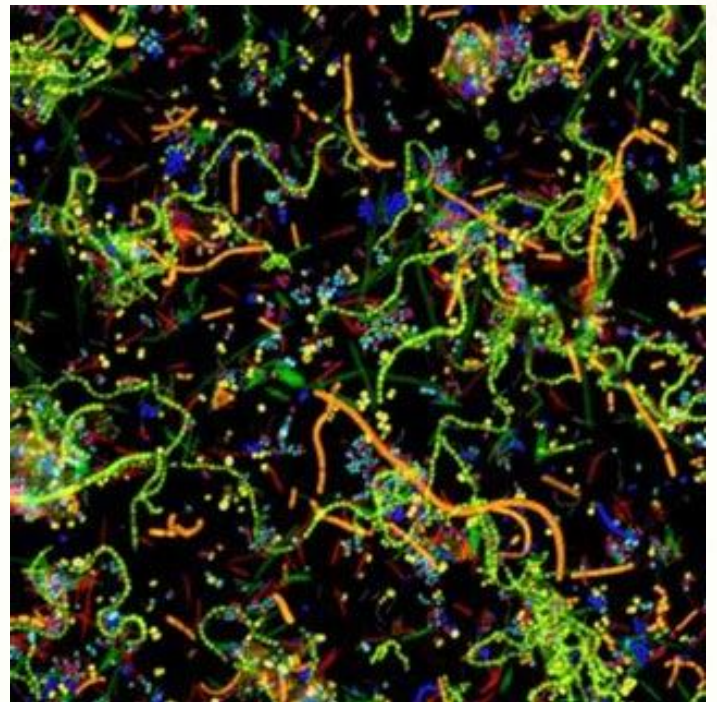


Oral 'animacules' (1676)



A. van Leeuwenhoek

Oral microbiome (2011)



Valm et al., PNAS

Isolate



Genomics / Transcriptomics

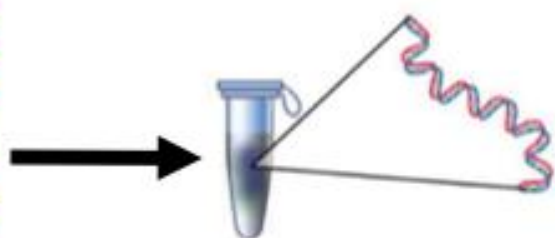
Community



Metagenomics / Metatranscriptomics



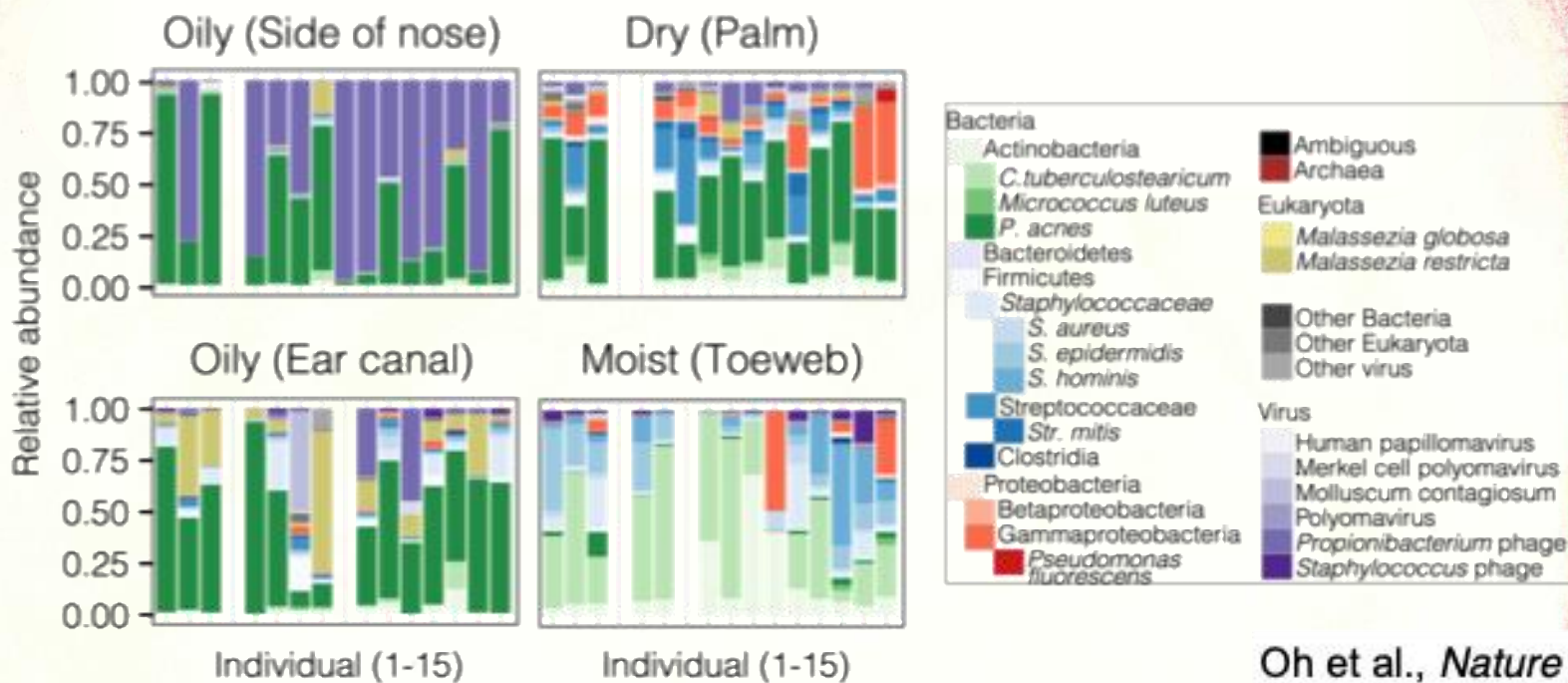
Microbial Community



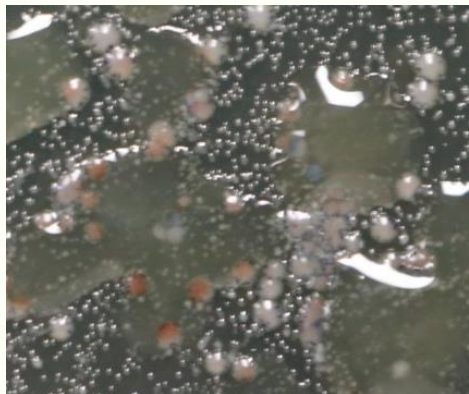
Molecular Extraction



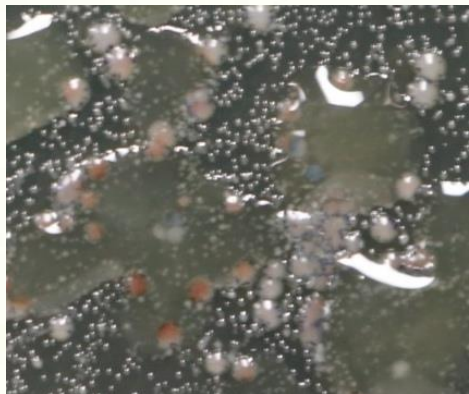
Sequencing



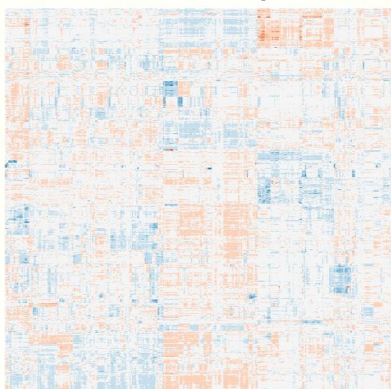
Microbial Culture



Microbial Culture

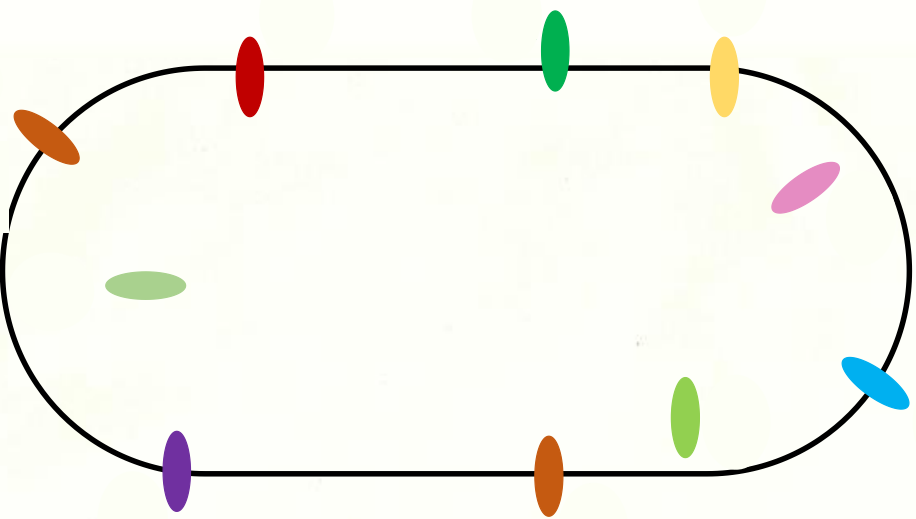


RNA-seq Data



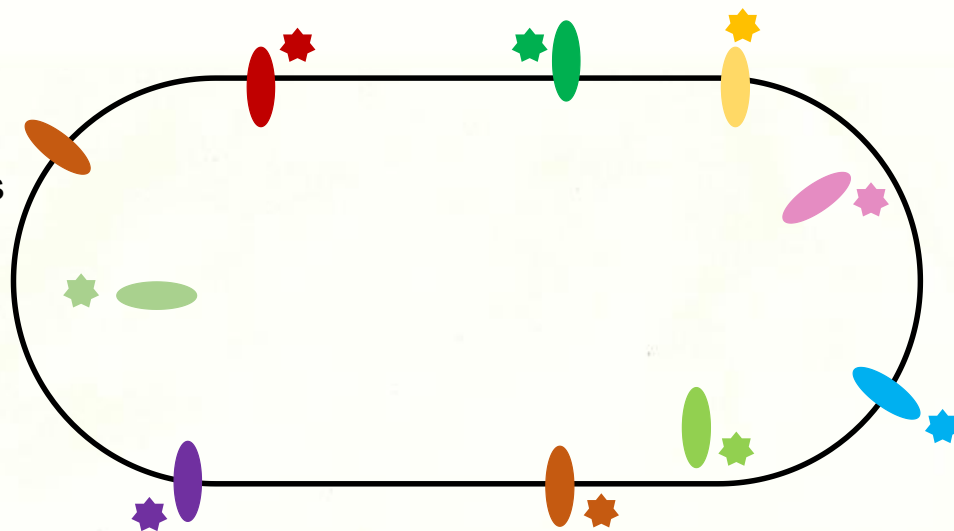


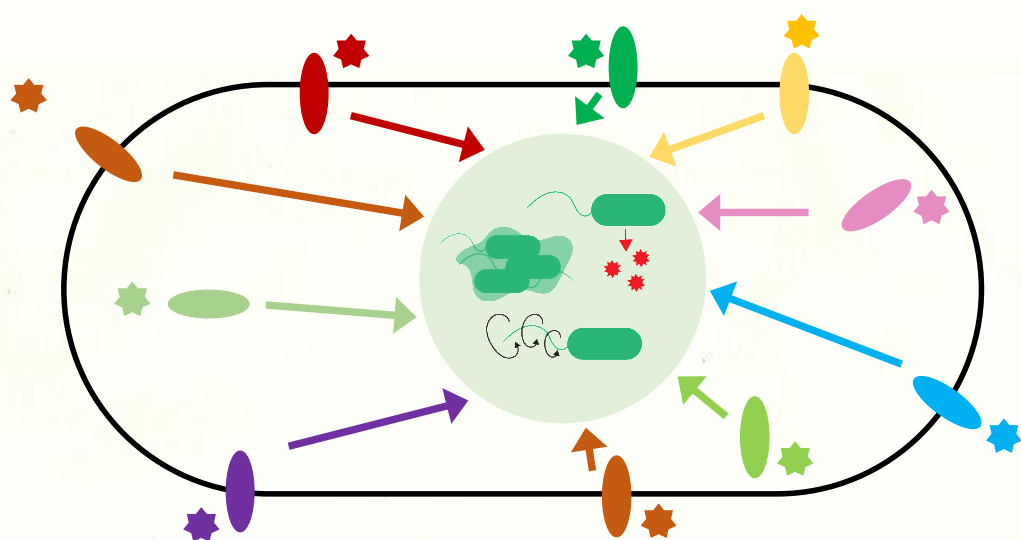
Proteins
sensors

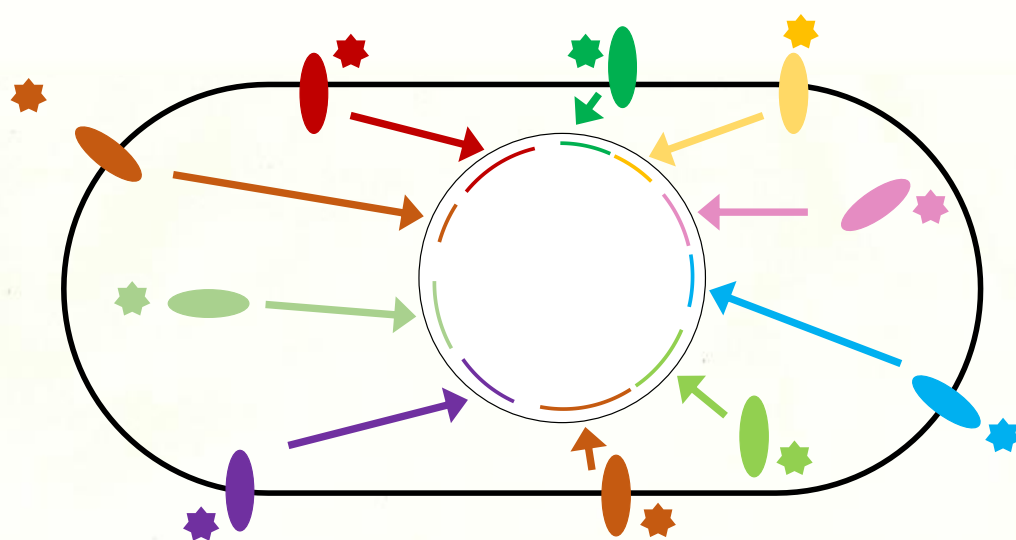


Molecular cues

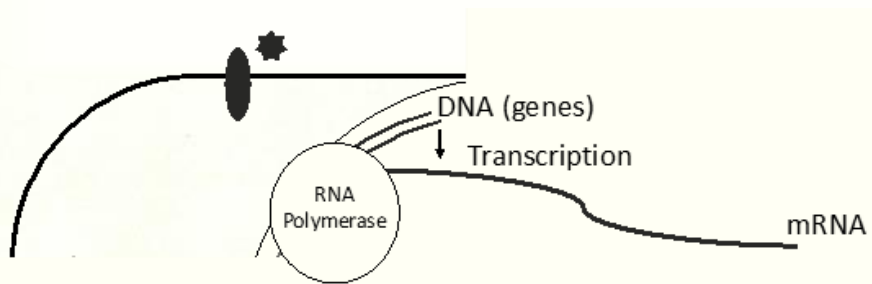
Proteins sensors

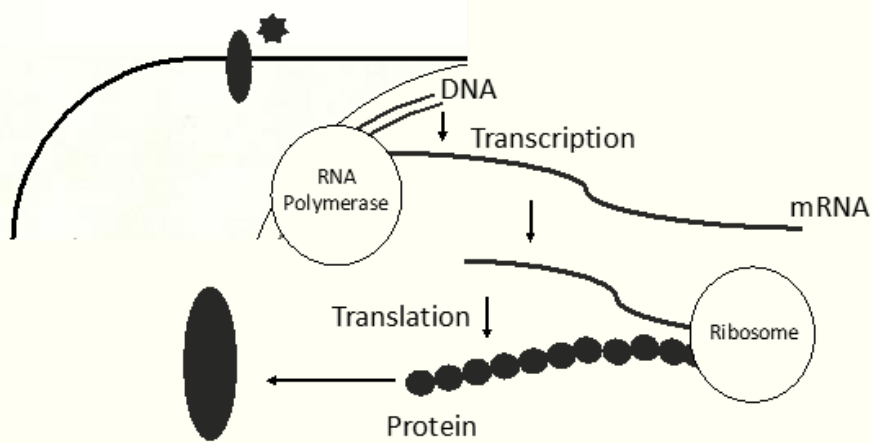


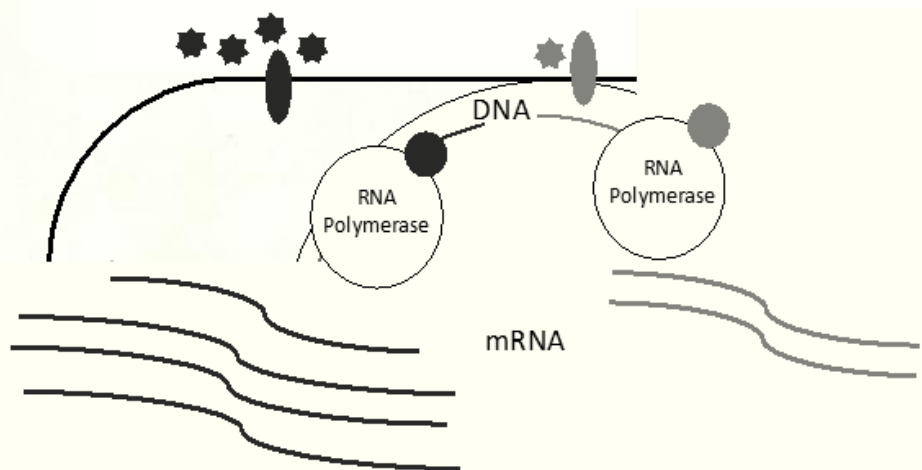


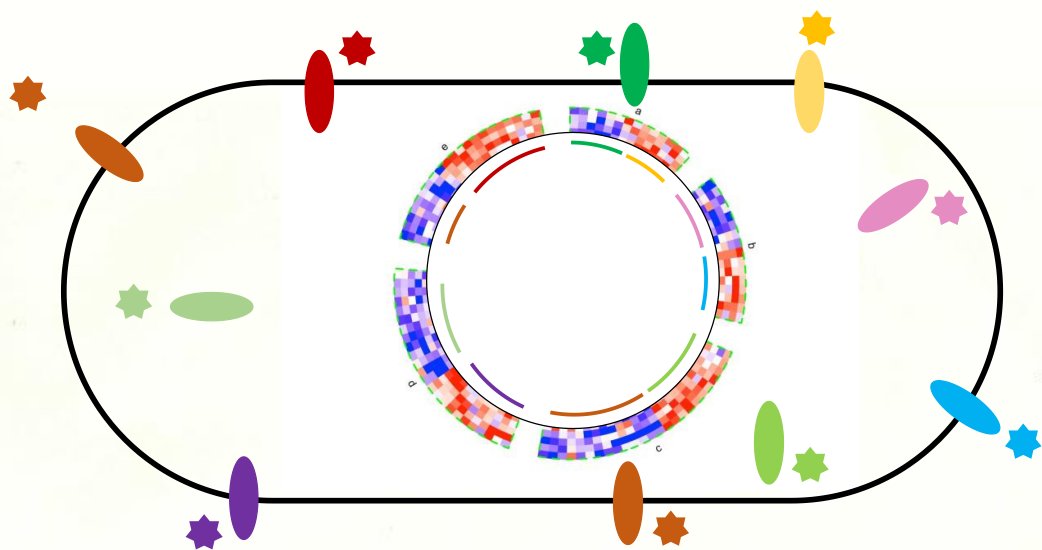


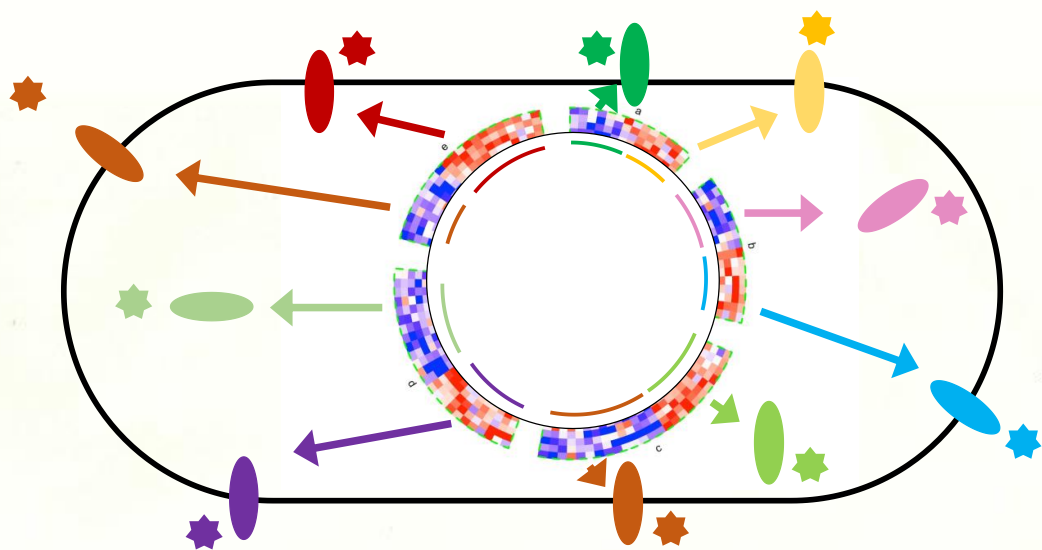


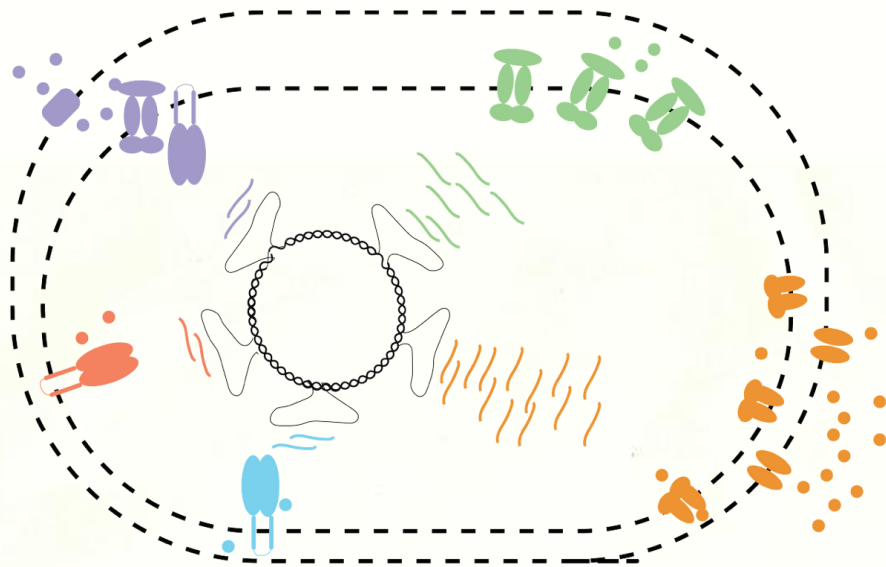


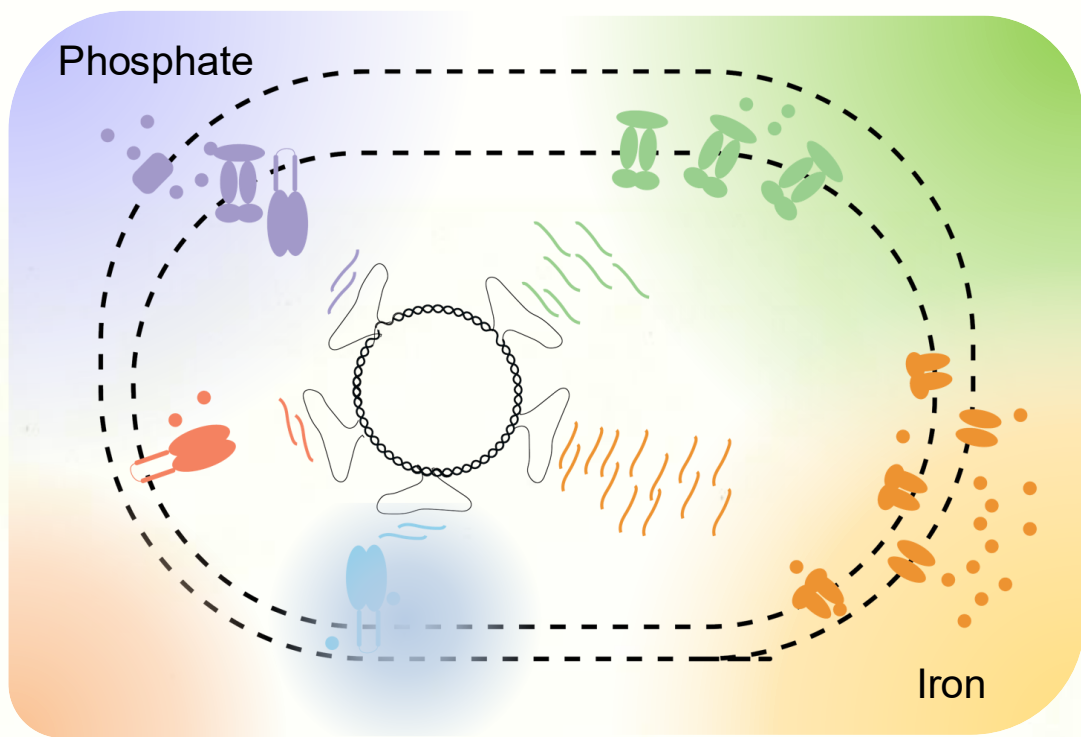


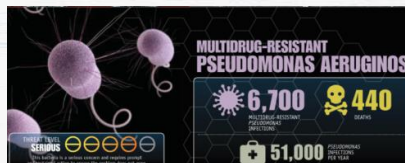
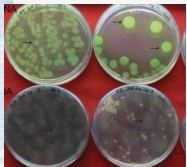
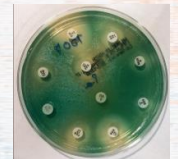
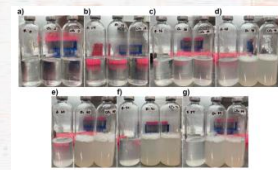
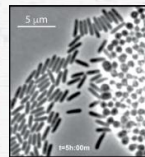
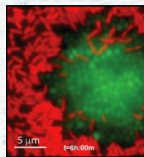
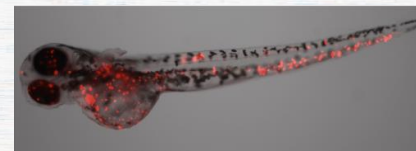
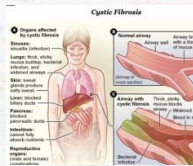
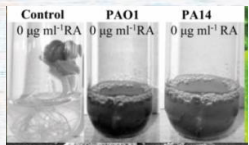


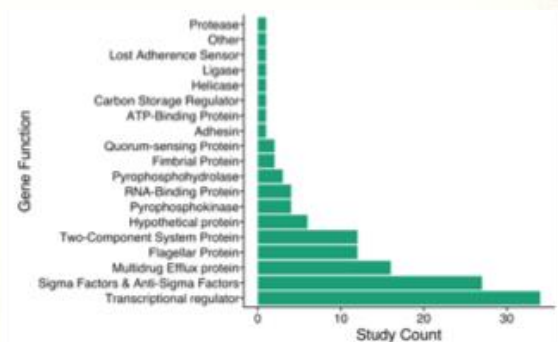


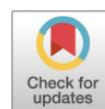












Computationally Efficient Assembly of *Pseudomonas aeruginosa* Gene Expression Compendia

 Georgia Doing,^a  Alexandra J. Lee,^b Samuel L. Neff,^a Taylor Reiter,^c Jacob D. Holt,^a Bruce A. Stanton,^a  Casey S. Greene,^{c,d}
 Deborah A. Hogan^a

^aDepartment of Microbiology and Immunology, Geisel School of Medicine at Dartmouth, Hanover, New Hampshire, USA

^bGenomics and Computational Biology Graduate Program, University of Pennsylvania, Philadelphia, Pennsylvania, USA

^cDepartment of Biochemistry and Molecular Genetics, University of Colorado School of Medicine, Denver, Colorado, USA

^dDepartment of Systems Pharmacology and Translational Therapeutics, Perelman School of Medicine, University of Pennsylvania, Philadelphia, Pennsylvania, USA

<https://pubmed.ncbi.nlm.nih.gov/36541761/>

ANTIMICROBIAL RESISTANCE THREATS

in the United States, 2021-2022

CDC used new data¹ to analyze the U.S. burden of the following antimicrobial-resistant pathogens typically found in healthcare settings:



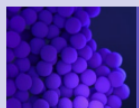
Carbapenem-resistant
Enterobacteriales (CRE)



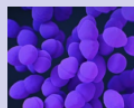
Carbapenem-resistant
Acinetobacter



Candida auris (*C. auris*)



Methicillin-resistant
Staphylococcus aureus
(MRSA)



Vancomycin-resistant
Enterococcus (VRE)



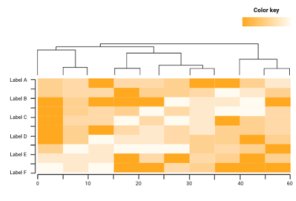
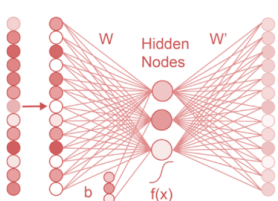
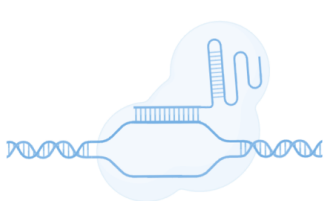
Extended-spectrum
beta-lactamase (ESBL)-
producing Enterobacteriales

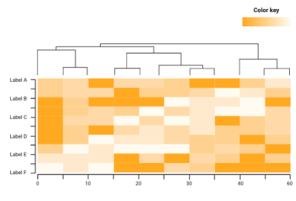
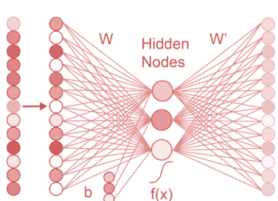
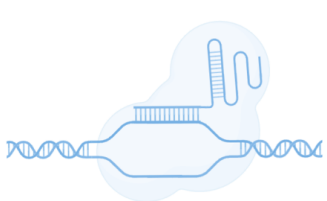


Multidrug-resistant (MDR)
Pseudomonas aeruginosa

CDC previously reported that the burden of these pathogens increased in the United States in 2020 in the [COVID-19 Impact Report](#). The information below describes the burden in the two following years, 2021 and 2022, and compares against 2019 data.

CDC <https://www.cdc.gov/ecoli/about/index.html>

SRA Search	Compendia	DAE Models	Bi-partite Graphs	Comparative Genetics
				
- organism: <i>E. coli</i> - molecule: mRNA - pull data from Short Read Archive (SRA) and Gene Expression Omnibus (GEO)	- metadata <i>E. coli</i> (20 strains) reference pan-genomes - salmon k-mer mapping (k=15)	- denoising autoencoder (tied) 1 layer of 50 hidden nodes - fully connected - objective): reconstruction error	$G(E,V)$ where $E \in \text{Corr}_{\text{gene, gene}}$ and $V \in \text{Genes}_{\text{S.e.}} \cup \text{Genes}_{\text{S.a.}}$ - Weight edges by correlation strength - Color genes by homology	- knock-down genes of interest under function-relevant conditions - look for fitness effects and/or phenotypes

SRA Search	Compendia	DAE Models	Bi-partite Graphs	Comparative Genetics
				
- organism: <i>E. coli</i> - molecule: DNA, RNA or text - pull data from Short Read Archive (SRA) and Gene Expression Omnibus (GEO)	- metadata - subset of strains and samples - reference pan- genomes - salmon k-mer mapping (k=15)	Dimensionality reduction: PCA ICA NMD DAE	Pattern detection: correlation regression clustering classification	Hypothesis testing: CRISPR data phenotypes (future directions)

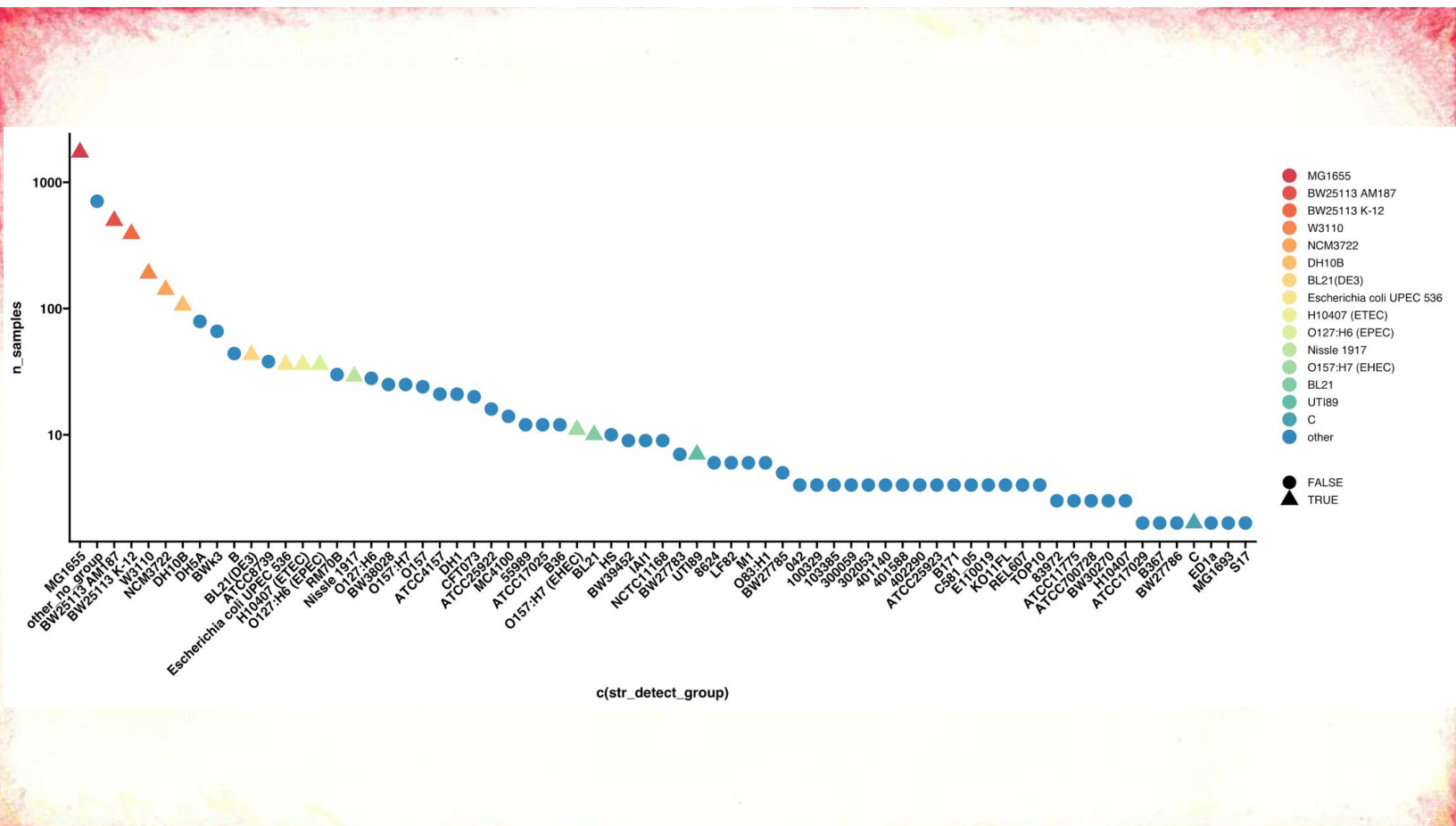
Ecoli Pangenome Compendium

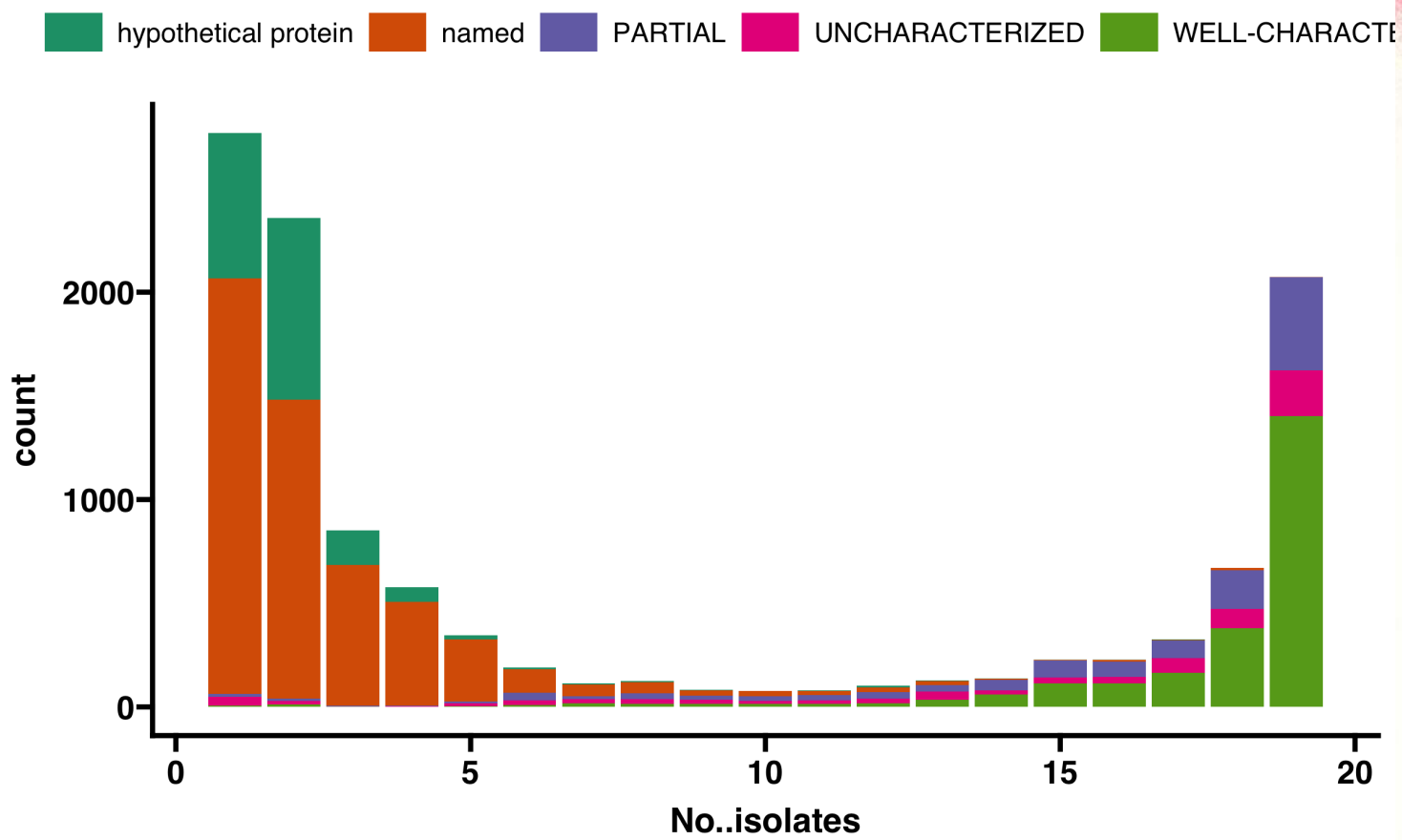
samples: 20,292

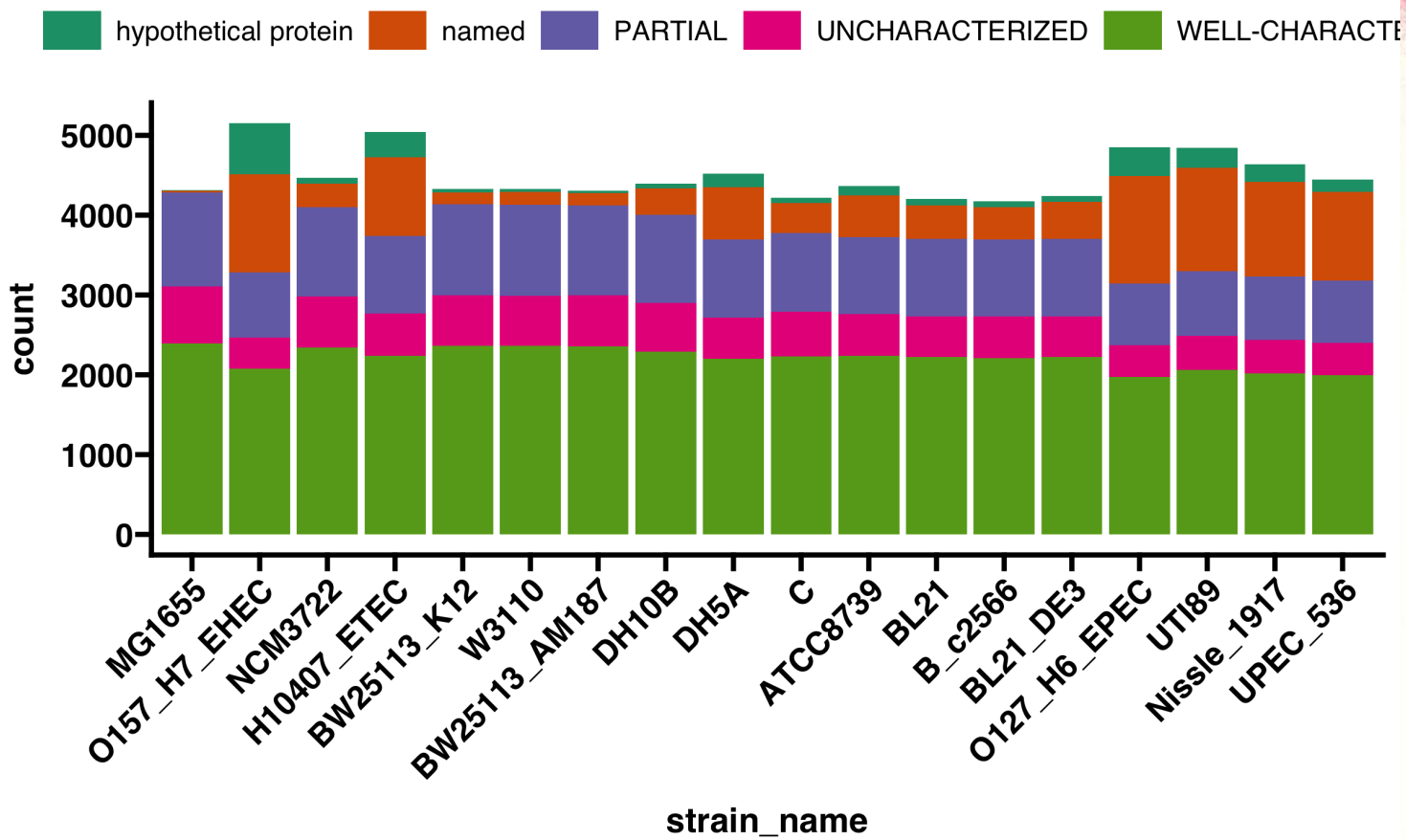
unique studies: 1,365

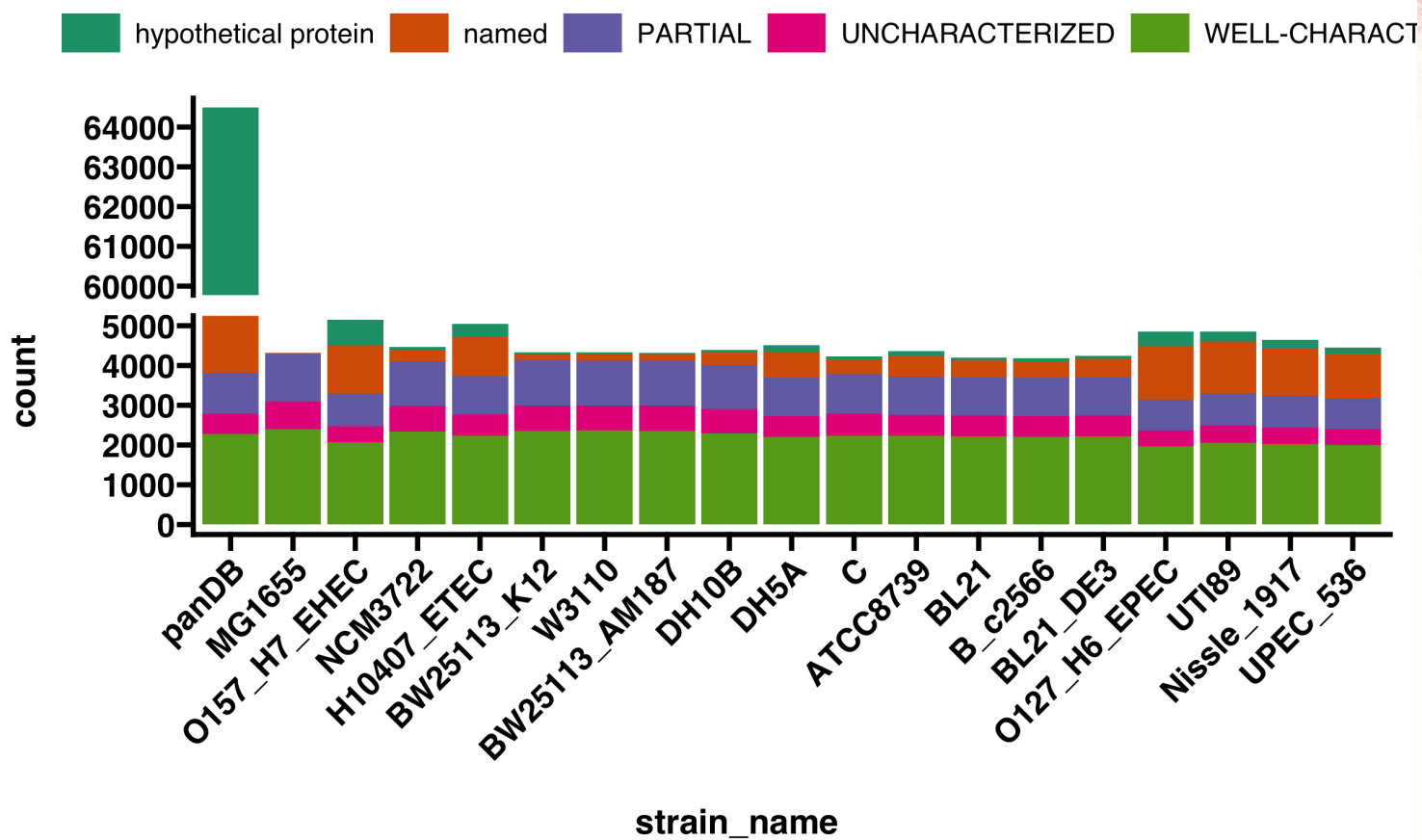
samples with high confidence strain: 4,260

samples with putative strain: 11,859



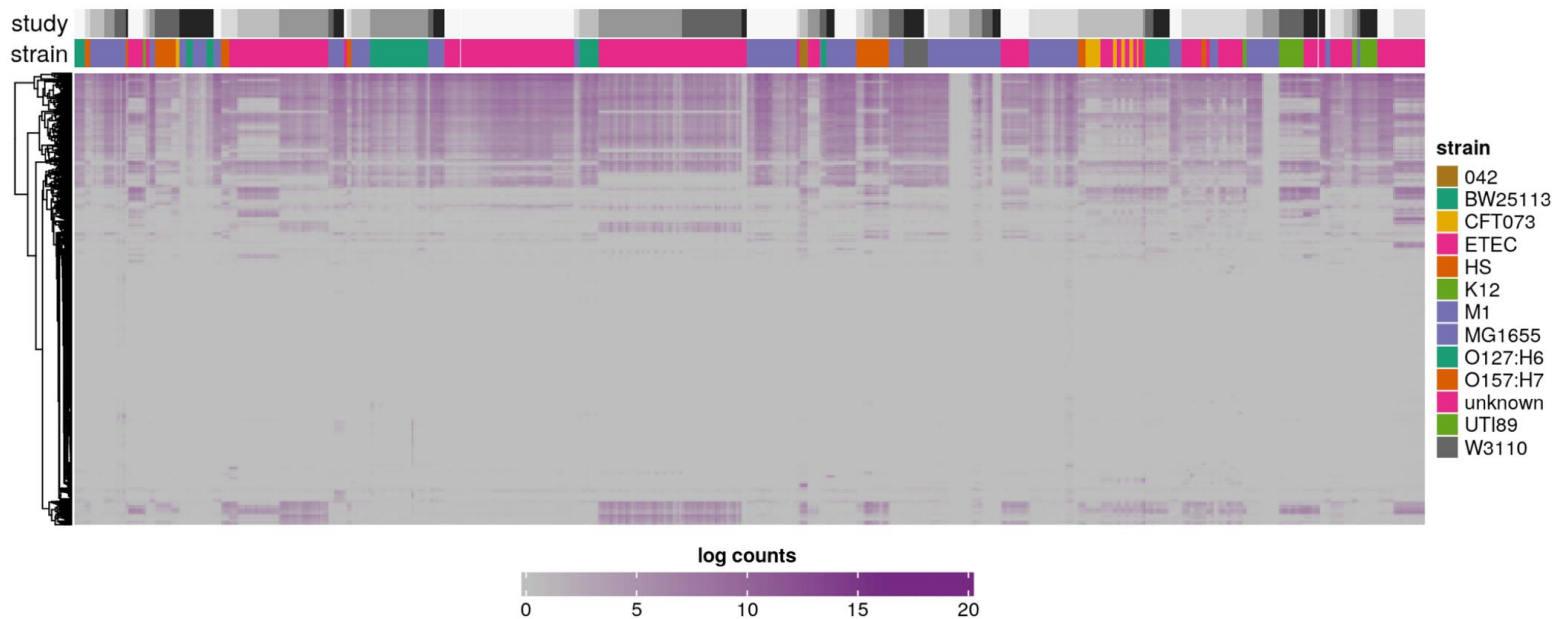


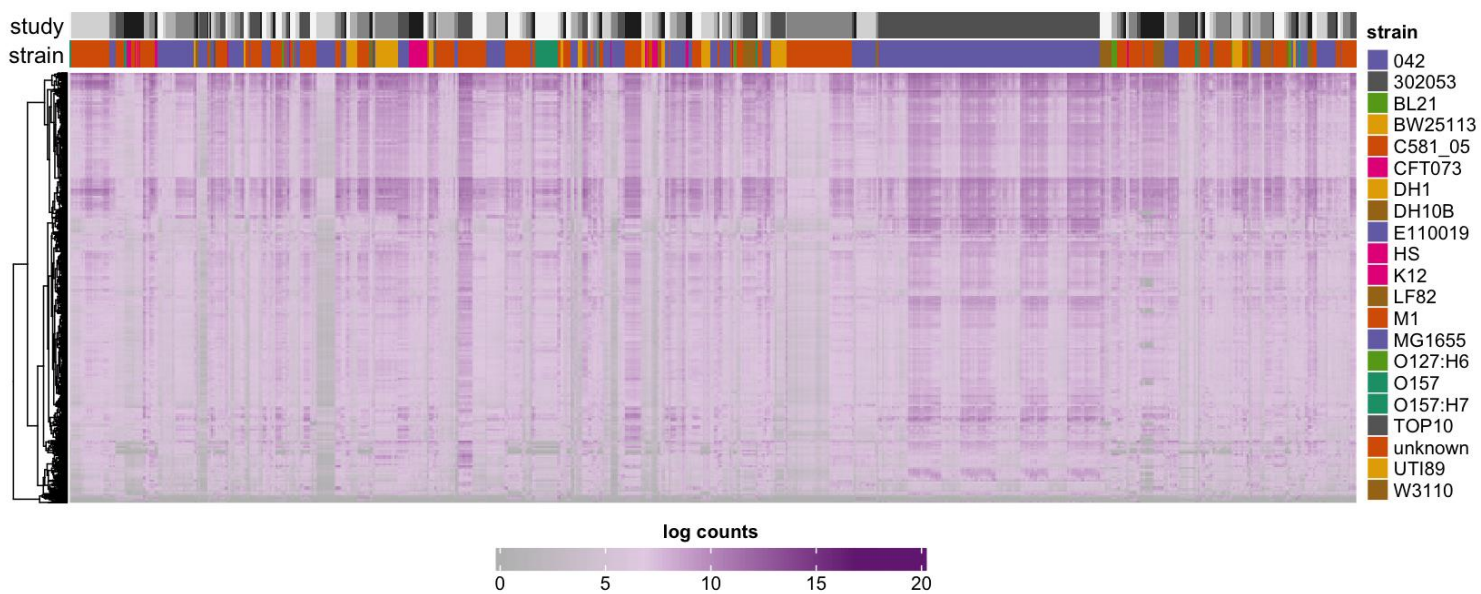




Head of “count” table

X	Name	ERX1805690	ERX1805691	ERX1805692	ERX2279850	ERX2279851	ERX2279852	ERX2279853	ERX2279854
0ed86576d343ce011a61773e0620e335_1045	character(0)	1.000	0.000	0.00	0.000	0.000	0.000	0.000	0.000
0ed86576d343ce011a61773e0620e335_1047	character(0)	0.000	0.000	0.00	0.000	0.000	0.000	0.000	0.000
0ed86576d343ce011a61773e0620e335_1120	character(0)	0.000	0.000	0.00	6.476	6.236	0.000	0.000	7.554
0ed86576d343ce011a61773e0620e335_1175	character(0)	0.000	0.000	0.00	2.000	2.000	3.000	2.000	0.000
0ed86576d343ce011a61773e0620e335_1179	character(0)	0.000	0.000	0.00	2.000	4.000	0.000	4.000	2.000
0ed86576d343ce011a61773e0620e335_1311	character(0)	0.000	0.000	0.00	0.000	0.000	0.000	0.000	0.000
0ed86576d343ce011a61773e0620e335_1450	character(0)	0.000	0.000	0.00	3.000	0.000	0.000	0.000	2.000
0ed86576d343ce011a61773e0620e335_1476	character(0)	5.403	2.351	0.00	15.370	11.212	23.317	5.232	12.820
0ed86576d343ce011a61773e0620e335_1484	character(0)	0.000	9.229	10.55	2.000	2.000	0.000	0.000	3.000
0ed86576d343ce011a61773e0620e335_2071	character(0)	0.000	0.000	0.00	0.000	0.000	36.005	0.000	0.000





From DEseq2, median of ratios normalization:

(Y_{ij}) : The raw count for gene (i) in sample (j) .

(g_i) : The geometric mean of counts for gene (i) across all samples.

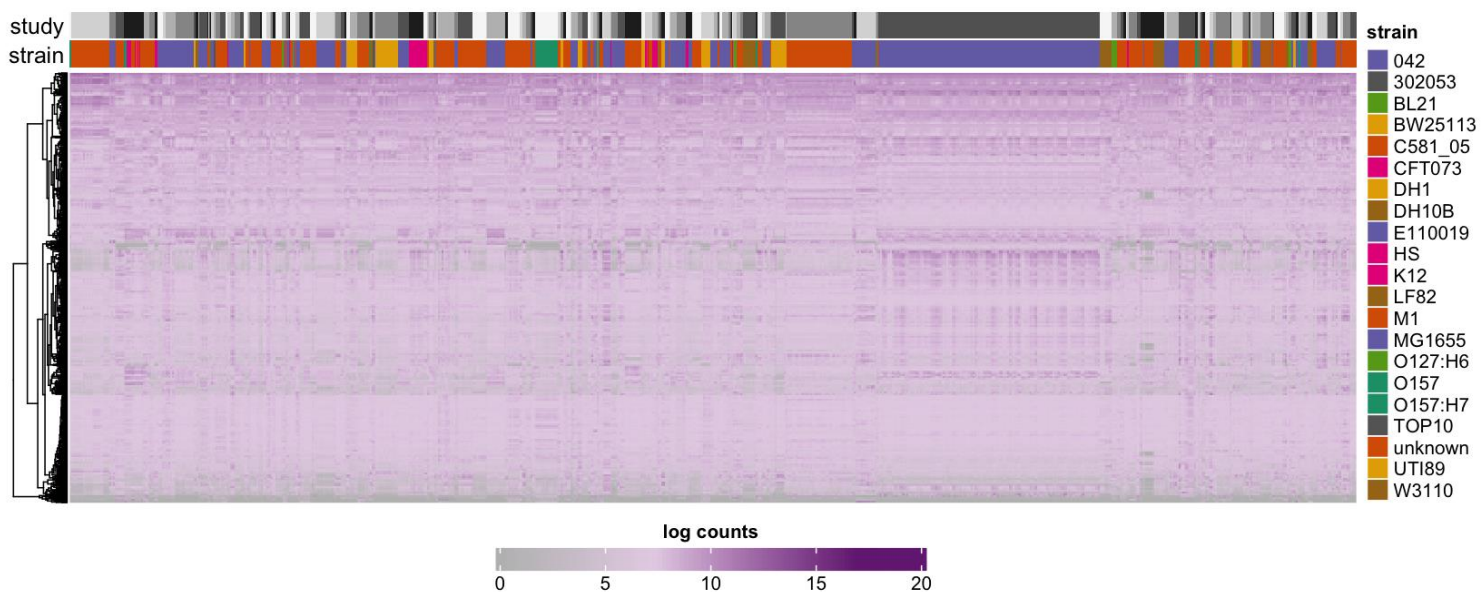
The size factor for sample (j) , denoted as (S_j) , is calculated as:

$$\left[S_j = \text{median} \left(\frac{Y_{ij}}{g_i} \right)_{i \in 1, \dots, m} \right]$$

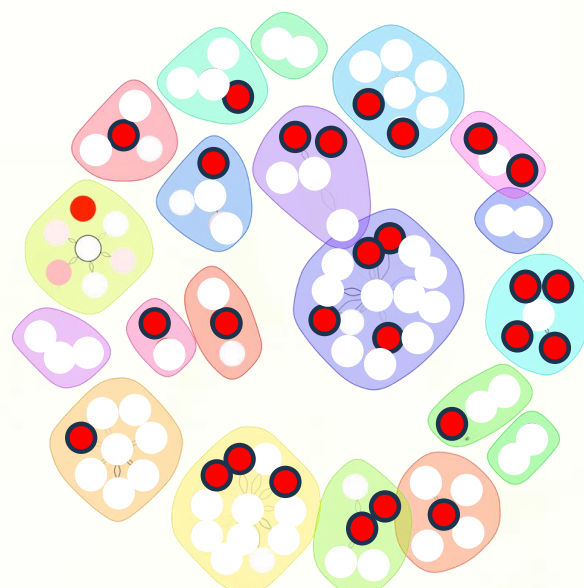
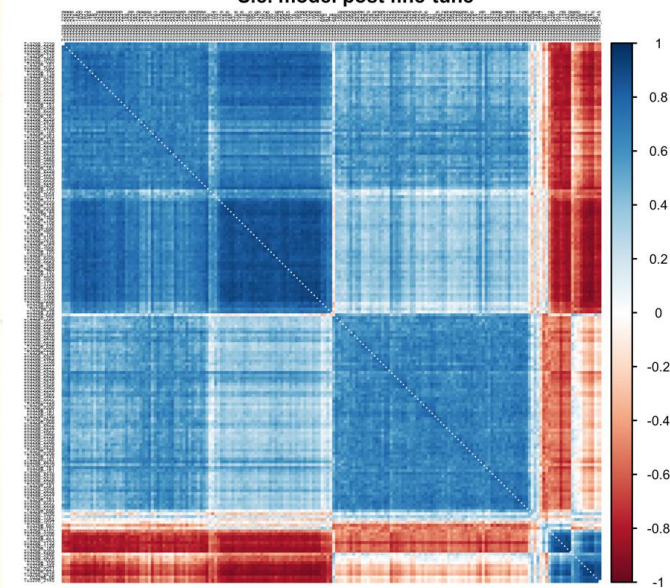
Where:

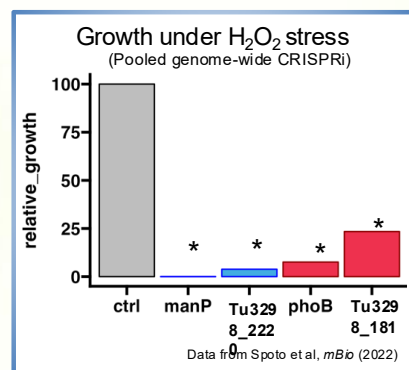
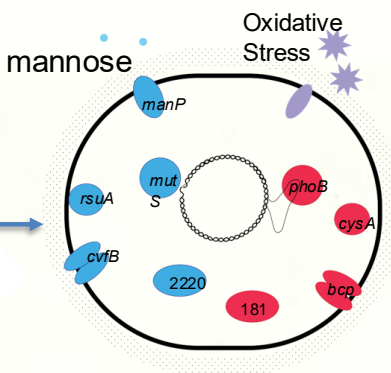
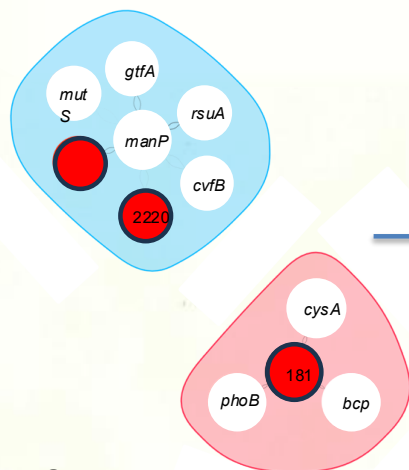
$(g_i = (\prod_{j=1}^n Y_{ij})^{1/n})$, the geometric mean of counts for gene (i) across (n) samples.

The ratio $(\frac{Y_{ij}}{g_i})$ is computed for each gene (i) in sample (j) . The median of these ratios is taken as the size factor (S_j) for sample (j) .



S.e. model post fine-tune





Lab machines: getting started

- Log on:
 - Your Union usernames
 - Default password: the word union followed by your ID number; e.g., union12345

Always log off before leaving the lab

Week 3

Alignment link

- alignment example:
- https://bioboot.github.io/bimm143_W20/class-material/nw/

E coli metadata activity

- colab notebook: https://colab.research.google.com/drive/1X18-sF7Spht9ZXZlDkY2S_zvgEEf4xJG?usp=sharing
- metadata 1: https://drive.google.com/file/d/1BB7M3eWo8zPuxmlwSKMzE_RBeYrv1n0/view?usp=sharing
- metadata 2: https://drive.google.com/file/d/18_PkR_sOLsQshSM4O1n2hVOcZoxgn8E0/view?usp=sharing
- metadata 3: https://drive.google.com/file/d/1_cJGV1gwHXzgUe7DMwHVI28nhJcma2tt/view?usp=sharing

Link to data processing reading

- original paper: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9948711/>
- review on gene expression compendia: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9396250/>

Notes on reading scientific paper

- from a previous Biological Data Science course at JAX
- colab notebook: https://drive.google.com/file/d/1V7MIeRD7Lp8a2hEwKD_qLd67Vk2mSc24/view?usp=sharing
- slides: <https://drive.google.com/file/d/19nR712iTWRr7S7HaNKmuG2BiU-psZY7/view?usp=sharing>

Khan Academy on the Central Dogma of Molecular Biology

- from AP biology course
- <https://www.khanacademy.org/science/ap-biology/gene-expression-and-regulation/translation/a/intro-to-gene-expression-central-dogma>
- chapters on gene expression, the genetic code, translation and more

Alignment Example

Syllabus

CSC 483 - Needleman-Wunsch Example

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2026-01-21

Intructions for the Needleman-Wunsch Alignment Algorithm

Given a mismatch scoring scheme $s(x_i, y_j)$ where x_i and y_j are nucleotides (nt) and g is a gap penalty, the optimal alignmnet score of two nucleotide sequences X and Y , indexed by i and j respectively, is :

$$S(i, j) = \max \begin{cases} S(i-1, j-1) + s(x_i, y_j) & \text{(Match or Mismatch)} \\ S(i-1, j) + g & \text{(Gap in Y)} \\ S(i, j-1) + g & \text{(Gap in X)} \end{cases}$$

For example, setting $g = -2$ and

$$s(x_i, y_j) = \begin{cases} 1 & \text{if Match} \\ -1 & \text{if Mismatch} \end{cases}$$

fill each square as:

$$S(i, j) = \max \begin{cases} \begin{cases} S(i-1, j-1) + 1 & \text{(Match)} \\ S(i-1, j-1) - 1 & \text{(Mismatch)} \end{cases} \\ S(i-1, j) - 2 & \text{(Gap in Y)} \\ S(i, j-1) - 2 & \text{(Gap in X)} \end{cases}$$

Dynamic Programming Implementation

1. In the upper left corner, initiate with 0, then, move across columns and down rows. Fill in each square with:

- the maximum value given the values at the previous indices
- the previous indices from which the value builds (diagonal, left or up arrows)

	Seq X		A	G	T
Seq Y		$i = 0$	$i = 1$	$i = 2$	$i = 3$
	$j = 0$	0			
A	$j = 1$				
A	$j = 2$				
G	$j = 3$				
C	$j = 4$				final score

2. Once all squares are filled, **the bottom right corner has your best possible score.**

Example

Step 1. Fill in scores for all i and j

For $i = 1, j = 0$:

$$S(i, j) = \max \begin{cases} S(0, j-1) + 1 = NA \text{ (indices cannot be } < 0) & \text{(Match)} \\ S(0, j-1) - 1 = NA \text{ (indices cannot be } < 0) & \text{(Mismatch)} \\ S(0, 0) - 2 = S(0, 0) - 2 = -2 & \text{(Gap in Y)} \\ S(1, j-1) - 2 = NA \text{ (indices cannot be } < 0) & \text{(Gap in X)} \end{cases}$$

$S(1, 0) = -2$, from $S(0, 0) = S(i-1, j)$, so mark with a left arrow

	Seq X		A	G	T
Seq Y		$i = 0$	$i = 1$	$i = 2$	$i = 3$
	$j = 0$	0	← -2		
A	$j = 1$				
A	$j = 2$				
G	$j = 3$				
C	$j = 4$				

Next, for $i = 2, j = 0$:

$$S(i, j) = \max \begin{cases} \begin{cases} S(1, j-1) + 1 = NA \text{ (indices cannot be } < 0) & \text{(Match)} \\ S(1, j-1) - 1 = NA \text{ (indices cannot be } < 0) & \text{(Mismatch)} \end{cases} \\ S(1, 0) - 2 = -2 - 2 = -4 & \text{(Gap in Y)} \\ S(2, j-1) - 2 = NA \text{ (indices cannot be } < 0) & \text{(Gap in X)} \end{cases}$$

$S(2, 0) = -4$, from $S(1, 0) = S(i-1, j)$, so mark with a left arrow

	Seq X		A	G	T
Seq Y		$i = 0$	$i = 1$	$i = 2$	$i = 3$
	$j = 0$	0	← -2	← -4	
A	$j = 1$				
A	$j = 2$				
G	$j = 3$				
C	$j = 4$				

Note, that $i = 3, j = 0$ follows similarly, as above. Give it a try, I've filled it in below.

Next, for $i = 0, j = 1$:

$$S(i, j) = \max \begin{cases} \begin{cases} S(i-1, 0) + 1 = NA \text{ (indices cannot be } < 0) & \text{(Match)} \\ S(i-1, 0) - 1 = NA \text{ (indices cannot be } < 0) & \text{(Mismatch)} \end{cases} \\ S(i-1, 1) - 2 = NA \text{ (indices cannot be } < 0) & \text{(Gap in Y)} \\ S(0, 0) - 2 = 0 - 2 = -2 & \text{(Gap in X)} \end{cases}$$

$S(0, 1) = -2$ derived from $S(0, 0) = S(i, j-1)$ (up arrow)

	Seq X		A	G	T
Seq X		$i = 0$	$i = 1$	$i = 2$	$i = 3$
	$j = 0$	0	← -2	← -4	← -6
A	$j = 1$	↑ -2			
A	$j = 2$				
G	$j = 3$				
C	$j = 4$				

The rest of the column ($i = 0$) follows similarly.

To fill in the center squares, we start with $i = 1, j = 1$:

$$S(i, j) = \max \begin{cases} S(0, 0) + 1 = 0 + 1 = 1 & \text{(Match)} \\ S(0, 0) - 1 = NA \text{ (not a mismatch, } A=A) & \text{(Mismatch)} \\ S(0, 1) - 2 = -2 - 2 = -4 & \text{(Gap in } Y) \\ S(1, 0) - 2 = -2 - 2 = -4 & \text{(Gap in } X) \end{cases}$$

$S(1, 1) = 1$ from $S(0, 0) = S(i - 1, j - 1)$, so mark with a diagonal arrow

	Seq X		A	G	T
Seq Y		$i = 0$	$i = 1$	$i = 2$	$i = 3$
	$j = 0$	0	$\leftarrow -2$	$\leftarrow -4$	$\leftarrow -6$
A	$j = 1$	$\uparrow -2$	$\nearrow 1$		
A	$j = 2$	$\uparrow -4$			
G	$j = 3$	$\uparrow -6$			
C	$j = 4$	$\uparrow -8$			

Continuing, for $i = 2, j = 1$:

$$S(i, j) = \max \begin{cases} S(1, 0) + 1 = NA \text{ (not a match, } G \neq A) & \text{(Match)} \\ S(1, 0) - 1 = -2 - 1 = -3 & \text{(Mismatch)} \\ S(1, 1) - 2 = 1 - 2 = -1 & \text{(Gap in } Y) \\ S(2, 0) - 2 = -4 - 2 = -6 & \text{(Gap in } X) \end{cases}$$

$S(2, 1) = -1$ derived from $S(1, 1) = S(i - 1, j)$ (left arrow)

	Seq X		A	G	T
Seq Y		$i = 0$	$i = 1$	$i = 2$	$i = 3$
	$j = 0$	0	$\leftarrow -2$	$\leftarrow -4$	$\leftarrow -6$
A	$j = 1$	$\uparrow -2$	$\nearrow 1$	$\leftarrow -1$	
A	$j = 2$	$\uparrow -4$			
G	$j = 3$	$\uparrow -6$			
C	$j = 4$	$\uparrow -8$			

Something to look out for, continuing, for $i = 1, j = 2$:

$$S(i, j) = \max \begin{cases} \begin{cases} S(0, 1) + 1 = -2 + 1 = -1 & \text{(Match)} \\ S(0, 1) - 1 = NA & \text{(Mismatch)} \end{cases} \\ S(0, 2) - 2 = -4 - 2 = -6 & \text{(Gap in Y)} \\ S(1, 1) - 2 = 1 - 2 = -1 & \text{(Gap in X)} \end{cases}$$

$S(1, 2) = -1$ could be from $S(0, 1) = S(i - 1, j - 1)$ (diagonal arrow)...

OR....could be from $S(1, 1) = S(i, j - 1)$ (up arrow).

	Seq X		A	G	T
Seq Y		$i = 0$	$i = 1$	$i = 2$	$i = 3$
	$j = 0$	0	$\leftarrow -2$	$\leftarrow -4$	$\leftarrow -6$
A	$j = 1$	$\uparrow -2$	$\nwarrow 1$	$\leftarrow -1$	
A	$j = 2$	$\uparrow -4$	$\leftarrow \nwarrow -1$		
G	$j = 3$	$\uparrow -6$			
C	$j = 4$	$\uparrow -8$			

Step 2. Find the final score in the bottom, right cell

$$S(X, Y) = -1$$

	Seq X		A	G	T
Seq Y		$i = 0$	$i = 1$	$i = 2$	$i = 3$
	$j = 0$	0	$\leftarrow -2$	$\leftarrow -4$	$\leftarrow -6$
A	$j = 1$	$\uparrow -2$	$\nwarrow 1$	$\leftarrow -1$	$\leftarrow -3$
A	$j = 2$	$\uparrow -4$	$\leftarrow \nwarrow -1$	$\nwarrow 0$	$\leftarrow \nwarrow -12$
G	$j = 3$	$\uparrow -6$	$\uparrow -3$	$\nwarrow 0$	$\nwarrow -1$
C	$j = 4$	$\uparrow -8$	$\uparrow -5$	$\uparrow -2$	$\nwarrow -1$

Step 3. Backtrace following the arrows for the optimal alignment

- To figure out the alignment that leads to that best possible score, back-trace from the bottom right to the upper left, following the arrows of highest score attribution. Starting with the last nucleotide (3' end), pre-pend the nucleotides as you go.

- if diagonal arrow, record both nt at those indices (they may or may not match)

- if left arrow, record the nt from the top sequence (Seq X , i indices) and a gap in the bottom sequence (Seq Y , j indices)
- if up arrow, record the nt from the bottom sequence (Seq Y , j indices) and a gap in the top sequence (Seq X , i indices)

Continuing the example

Starting in the bottom right cell $i = 3, j = 4$, given the diagonal arrow, we begin our backward alignment by recording both nucleotides at these indices:

```
Seq X:    T
          |
Seq Y:    C
```

This brings us to $i = 2, j = 3$, with another diagonal arrow so both nt are added again:

```
Seq X:    GT
          ||
Seq Y:    GC
Match:    *
```

Then to $i = 1, j = 2$, with a diagonal arrow **and** a left arrow, so we have two options:

Option 1, diagonal arrow, add both nt:

```
Seq X:    AGT
          |||
Seq Y:    AGC
Match:    **
```

Option 2, up arrow, add a gap to Seq2 (i indices):

```
Seq X:    -GT
          |||
Seq Y:    AGC
Match:    *
```

Finishing out the trace, we can choose an option (maybe follow the arrow that points to the higher score) or we can follow both options:

Option 1, now we are at $i = 0, j = 1$ with an up arrow so add a gap to Seq2:

```
Seq X:    -AGT
          ||||
Seq Y:    AAGC
Match:    **
```

Option 2, now we are at $i = 1, j = 1$ with a diagonal arrow so both nt get added:

```

Seq X:  A-GT
        ||||
Seq Y:  AAGC
Match:  *  *

```

Note that both alignments produces a score of -1 and both have 2 mismatches. If we broke this tie by following the path that always chooses the higher score, we would end with Option 2 since $[S(1, 1) = 1] > [S(0, 1) = -2]$

You can see this using the online tool by setting your sequences to

```

Seq1 (Y): AAGC
Seq2 (X): AGT

```

https://bioboot.github.io/bimm143_W20/class-material/nw/

If you "Clear Path" and manually trace Option 1, however, you will get the same score.

Practice

Try this again with new sequences (any length), check your results using the online tool.

	Seq X									
Seq Y		$i = 0$	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$	$i = 7$	$i = 8$
	$j = 0$									
	$j = 1$									
	$j = 2$									
	$j = 3$									
	$j = 4$									
	$j = 5$									
	$j = 6$									
	$j = 7$									
	$j = 8$									

Alignment Example

Syllabus

about.qmd

References

